JAOS 109 (1989) 183-203 Bib 33

THE TIBERIAN REFLEXES OF SHORT *i IN CLOSED SYLLABLES

E. J. REVELL

UNIVERSITY OF TORONTO

The Tiberian Hebrew reflex of original short *i is not always that expected under the accepted rules of sound change, such as Philippi's law. This has generally led to the opinion that, in the case of this vowel, the standard process of change described by such rules has often been disrupted, for instance by analogic levelling. This article argues that, if a wide enough range of conditioning factors is taken into consideration, the Tiberian reflexes can, in the majority of cases, be shown to result from a natural process of change. The phonetic environment, particularly the consonant preceding the vowel, has scarcely been considered before as a factor conditioning the development of *i. Its significance is demonstrated here. The reflexes of the vowel in a representative selection of forms are related in detail to this and other conditioning factors. An attempt is then made to describe the process of sound change which gave rise to the various Tiberian reflexes.

INTRODUCTION

1. THE ORIGINAL SHORT *i[1] of a common Semitic or Proto-Hebrew form may give rise, in the Tiberian text of the Bible (MT), to any one of five of the seven vowels of the standard Tiberian system. This diversity has generally led students of the history of the language to the opinion that the natural process of linguistic change has, in the case of this vowel, often been disrupted (see, e.g., Bauer-Leander 1922, §14.a′; Lambdin 1985, 144-45). The present article is an attempt to show that such disruption occurred much less frequently than is generally suggested—that is, that in most cases the appearance of a particular MT reflex of *i can be ascribed to particular phonological conditioning.

2. This study is limited to the common cases in which *i was the original vowel of a syllable which is stressed and closed (or secondarily opened) in an MT base form. The reflexes of *i in both base and inflected forms of such words are studied. The following forms are included.

NOUNS—of forms *CiC, as *ben*, *CiC:, as *ʾem*, *CiCC, as *s'efɛr*, *CaCiC, as *zɔq'en*, a few other noun forms of two syllables or more showing *ṣere* before the final root letter where the syllable is stressed, closed and final in MT, as *mɔg'en*, or *ḥireq* where it is unstressed, closed, and non-final, as *mor'ag*, *morig:'îm*.

—Participles *qal* and *piʿel*.

—Infinitives *nifʿal*, *piʿel* and *hifʿil*.

VERBS—*Qal* perfect of form *qatil (see §2.1), as *kɔb'ed*, imperfects from roots III*h*, and from initial

[1] In transliterations, the Tiberian vowels are represented by *u*, *o*, *ɔ*, *a*, *ɛ*, *e*, *i*, *shewa* by *ə*, *ɔ̆*, *ă*, *ɛ̆*. The use of *w* or *y* as a vowel letter with *u*, *o*, *e*, or *i* is indicated by a circumflex over the vowel. Elsewhere, *h*, *w*, and *y* which do not represent consonants are ignored. Stress position is indicated by a vertical stroke before the vowel of the syllable. The lengthening of a consonant is indicated by a following colon. Capital letters are used to indicate the consonants of a root. In the descriptions of vowel change, the vowels of the received text (MT) are indicated by name. A reconstructed "original" vowel—that from which the MT vowel is believed to derive—is marked with an asterisk, as *i, *a. Other vowels are indicated by the appropriate letter italicized. Thus in the stressed syllable of *hiqt'altɔ*, *i has given rise to *a*, MT *pataḥ*.

This paper has been read by Prof. S. Morag and Prof. J. W. Wevers; I am grateful to both for helpful criticism.

The writer's copy of E. Qimron's article, "Interchanges of *e* and *a* Vowels in Accented Closed Syllables in Biblical Hebrew," *Leshonenu* 50 (1985-86): 77-102, arrived only after this article had been accepted and revisions submitted. It would no doubt have been improved by attention to the points raised by Qimron, but no substantial changes would have been made. The crucial factor in which the present approach differs from that of Qimron is its unwillingness to accept that change due to analogy should have been so widespread and yet have such random effects.

weak roots producing imperfect forms like *yoʾk'al*, *yeš'eb*, *yit:'en*, *yɔš'îm*.[2]

—*Nifʿal* imperfects and imperatives.

—*Piʿel* perfects, imperfects and imperatives.

—*Hifʿil* perfects, imperfects and imperatives in which the vowel before the last root consonant is other than *ḥireq*. Information on other forms is included, but not consistently.

2.1 *Qal* perfects are identified as reflecting the pattern *qatil if the vowel before the last root consonant is represented as any one of the following:

(i) *Ṣere* where the syllable is closed stressed and final in context or in pause,[3] as *ʾɔh'eb*.

(ii) *Pataḥ* where the syllable is closed, stressed and non-final in pause, as *zɔq'antî* (Gen 18:13, with *silluq*).

(iii) *Ṣere* where the syllable is open, stressed or unstressed, as *ʾăheb'ô* (Gen 44:20). The relation between the patterns of the perfect and imperfect *qal* in any root, *qɔt'al* with *yiqt'ol* and *qɔt'el* with *yiqt'al*, is not fully consistent (cf. ḤPṢ, ŠKN). Consequently the use of an imperfect of pattern *yiqtal* is not considered to be evidence that the perfect reflects *qatil (cf. ŠKB).

These criteria identify the following roots as reflecting the pattern *qatil in the *qal* perfect: ʾHB, ʾŠM, BṢQ, GBR, GDL, DBQ, DŠN, ZQN, ḤDL, ḤMṢ, ḤNP, ḤPṢ, ḤPR, ṬHR, YBŠ, YʿP, YRŠ, YŠN, KBD, KŠR, LBŠ, NʿM, ʿṢM, ʿŠN, ʿŠŠ, QDŠ, QML, QRB, RʿB, ŚMḤ, ŚʾB, ŠʾL, ŠKḤ, ŠKN, ŠMN, ŠMʿ, ŠPL. The distinction of nominal and verbal forms generally follows that of the *Dictionary* of Brown, Driver and Briggs.

2.2 The vowel between the last two root consonants of *hitpaʿel* verbal forms was evidently a*, as the MT reflex is invariably *qameṣ* in pause, not only in stressed syllables, whether open or closed, final or non-final, but even in the unstressed open syllable in forms such as *titḥam:ɔq'în* (Jer 31:22). Consequently these forms are not included, despite the appearance of *ṣere* (and occasionally *ḥireq*) in contextual position. In nominal forms (participle, infinitive) the vowel of a closed stressed final syllable is uniformly *ṣere* in pause or in context, showing that the original vowel is *i. Consequently, forms of interest are noted where relevant. The only anomaly appears to be the *qameṣ* in the penultimate open unstressed syllable in *mitqô-mɔm'ɔ* (Job 20:27).

3. Original *i gives rise to any one of five different MT reflexes in these forms, because several conditioning factors are involved, and individual factors may be involved to a different degree in different situations. The effect of one factor may modify that of another, so that, where each factor is described individually, none appears to produce consistent effects. However a clear demonstration of the influence of the individual factors requires that each be described individually as far as is possible. As a compromise, the factors conditioning the development of *i are treated here under the following headings:

The structure and position of the syllable (§§6–11). The present study is concerned only with syllables which are closed in MT, or which were originally closed, but are open in MT. The development of *i may differ in the different types. The development of *i in such syllables was also affected by the position of the syllable in the word (a) in relation to other syllables, or (b) in relation to the stress pattern. The syllable may be final (not followed by other syllables), internal (both preceded and followed by other syllables), or initial (not preceded by other syllables in the word in isolation—the vowel of an initial syllable preceded by a prefix typically does not act as does the vowel of an internal syllable). A syllable is regarded as stressed if it is marked with the main accent of the word in MT (or, where the accent is preposed or postposed, if it is potentially so marked). Other syllables are regarded as unstressed. Words marked with the hyphen *maqqef* have no stressed syllable. These factors have been considered on much the same basis in former studies.

Phonetic Environment (§§12–19). The development of *i is affected by the nature of the neighboring sounds. In most previous studies the only feature of the phonetic environment considered is the following consonant, classified as "guttural" or "other." The influence of the following consonant is considered in greater detail in Rabin 1968. The present study shows that the development of *i is affected by the preceding consonant as well. It may also be affected by neighboring vowels.

Relation to Context (§§20–25). The development of *i is affected by the position of the word in which it stands in the larger spoken unit of which it forms a part. In former studies, this category has typically been seen in terms of the form of that word as contextual or pausal, or as construct or absolute. The former variation marks the word as medial or termi-

[2] Whatever the earliest form of the vowel between the two root consonants of a hollow root form, or the last two in a *hifʿil* form, in closed syllables it acts in biblical Hebrew as do originally short vowels.

[3] The forms from BŠL, ḤRD, ŠLM, ŠMN, are considered nominal.

nal in one of the units into which the text was divided in its spoken realization. The latter variation represents one facet of the syntactic/semantic relationships on which this division was based. The accents are traditionally regarded as marking these divisions and their subdivisions. Relationship between the accentuation and the reflexes of *i was noted in Rabin 1968. However the use of about 20% of pausal forms (which mark the ends of major units) cannot be predicted on the basis of the accents, so it is clear that the accents do not provide a fully accurate indication of the boundaries of such units. The relation between the accent system and those subdivisions of major units which are reflected by such vocalic variation is even less clear. The boundaries of these subdivisions can be predicted (with unknown accuracy) on the basis of syntactic relationships other than construct, and of such semantic relationships as can be objectively described.[4]

Other Factors (§§26–29). It can be shown that most, but not all, of the MT reflexes of *i could have resulted from natural sound change conditioned by the factors listed above. Other factors are suggested which may possibly have accounted for the main groups of exceptional reflexes.

4. It seems probable that vowel length was a significant factor in the vowel changes to be studied. The five MT reflexes of *i are often assigned to two (or three) classes characterized by difference in length. However difference in length is neither marked in the text, nor described by early writers. Moreover it is difficult to believe that the generally accepted views are well founded. The statement that *pataḥ* is "normally a short vowel" appears incompatible with the statement that "vowel length is determined by the position of the vowel within the word," since *pataḥ* can appear in the same positions as any other vowel.[5] Consequently it must be accepted that there is no reliable data on vowel length. Any description of this factor must appear among the conclusions of the study.

5. Original *i may be reflected in MT by *ḥireq*, *ṣere*, *segol*, *pataḥ*, and *qameṣ*. In this order, the vowels may be considered as progressing from the sound closest in quality to *i* to that most distant, a view of the relationship of the Tiberian vowels which can be justified from the statements of early masoretes and grammarians.[6] Following this order, the appearance of *pataḥ* as MT reflex of *i is considered to result from greater change than the appearance of *ṣere*. Consequently the conditions which give rise to *pataḥ* are said to be "conducive," or "favorable" to change; those which give rise to *ṣere* are "unfavorable." Conditions which give rise to *qameṣ*, *segol*, or *ḥireq* are respectively "most favorable," "moderately favorable," or "least favorable" according to the position of the vowel in the hierarchy.

THE STRUCTURE AND POSITION OF THE SYLLABLE

6. *Syllables originally closed and non-final which are non-final or doubly closed in MT.* This section includes (i) syllables which have always been closed and non-final, and which are either stressed, or occur in an inflected form of a word in which they are stressed in the base form, and (ii) syllables which were originally closed and non-final, but which are, in MT, either doubly closed and final, or open and non-final due to segolation (the breaking of the double closure by an anaptyctic vowel). Short vowels in this second group evidently developed in the same way as in closed stressed non-final syllables through most of their history, as is shown in Revell, 1985a. Any of the possible reflexes of *i may appear in one of these closed non-final syllables. Stress level is clearly a conditioning factor. *Qameṣ* occurs only in pausal position; *ḥireq* almost only in unstressed syllables. However although *pataḥ* is most common where the

[4] For the background to these views, see Revell 1985b, §§5–13; 1987a, §§1.15–17, 2.49–56.

[5] The quotations are from Moscati 1964, §8.80, p. 50. However note the parallels between *qameṣ* and *pataḥ* in stressed syllables, as *wat:'ahar* (Gen 16:4), *way:'ɔḥal* (2 Kgs 1:2); *wɔm'at:î* (Gen 19:19), *wɔm'ɔtnû* (Deut 5:21)—all pausal, and cf. *b'ayit* with *m'ɔwɛt*. The principal difference in such cases must be one of quality, not one of length. Masoretic treatises state that a vocal *shewa* before a guttural has the quality of the vowel following the guttural, e.g., in Levy 1936, text p. 18f., translation p. 17*. Thus any vowel quality may appear in *shewa* length. Since this paper was submitted for publication, an extremely interesting study of vowel length has appeared in Kahn 1987.

[6] The earliest securely datable description is that of Saadya Gaon, published in Skoss 1952, 292. This work dates from about 925, the end of the period of the Masoretes. The most exact description is that of the *Hidāyat al-Qāri*, compiled about a century later, published in Eldar 1985, 242. Some masoretic sources, such as *Okhlah we-Okhlah* list §5, probably antedate the development of vowel signs, as remarked in Díaz-Esteban 1975, 21. These demonstrate the relationship between *pataḥ* and *segol*, *ṣere*, and *ḥireq* but not that between the last three named.

syllable is stressed, it is also common in unstressed syllables, while *ḥireq*, *ṣere*, and *segol* all occur under stress. Evidently, then, the importance of stress as a conditioning factor has been over-emphasized. Other conditioning factors are the position of the syllable in the word and the nature of its closure. Change was evidently inhibited where the syllable was word-initial (disregarding prefixed particles) or where the vowel was followed by a lengthened consonant. Consequently *ḥireq*, *ṣere*, and *segol* often appear in these positions. However *pataḥ* is the most common reflex of *i in closed non-final syllables generally, so it is here taken to be the expected reflex. The description below is directed to the cases in which *pataḥ* does not occur.

6.1 *Ḥireq* occurs most commonly where the syllable is unstressed, and is either word-initial, or is closed by a lengthened consonant. Thus:

(i) *Ḥireq* occurs in a word-initial syllable in most two-syllable segolate forms, as *(lə)lidt'î* (1 Kgs 3:18), also in nouns of form *CiC:, where the base-initial syllable is closed by a lengthened consonant, as *bit:'î*. *Pataḥ* occurs in this position in forms from *kan*, *mad*, *gan*, and in a few segolate forms (see Revell 1985a, §2, 3, 5). *Ḥireq* occurs under stress in a word-initial syllable (so far as I have observed) only in short imperfect forms from roots III*h*, commonly where the syllable has been secondarily opened, as *way:'ibɛn*, but also where it has remained closed, in *way:'ipt* (Job 31:27) and *way:'išb* (Num 21:1, Jer 41:10).

(ii) *Ḥireq* occurs before a lengthened consonant in an unstressed syllable in a few noun forms, as *ʾămit:'ô* (Ps 91:4), *məbaʿit:'ɛkɔ* (1 Sam 16:15), and commonly in verb forms, as *haḥit:'otɔ* (Isa 9:3). However where the lengthening of the consonant results from assimilation of the final root consonant to the pronominal affix of a perfect verb form, the reflex of *i is regularly *pataḥ*, as *wəhišbat:'ɛm* (Exod 5:5). An exception occurs in *hămit:'ɛm* (Num 17:6).

(iii) Under other conditions, the reflex of *i in a closed non-final syllable is regularly *pataḥ* in verb forms, whether the syllable is stressed or unstressed, as *ʾɔh'abtɔ*, *ʾăhabt'ɛm*. *Ḥireq* occurs only in *šəʾilt'îhû* (Judg 13:6), *šəʾilt'îw* (1 Sam 1:20) *hišʾilt'ıhû* (1 Sam 1:28), and in all forms from YRŠ, as *wîrišt'ɛm* (Deut 4:1), *wîrišt'ɔh* (Deut 17:14), *wîrišt'ɔm* (Deut 19:1), except in the *waw* consecutive form *wəyɔrašt'ɔ* (Deut 6:18). There are several examples of all four forms. *Pataḥ* is also regularly the reflex in unstressed syllables in *qal* participles, as *yôladtək'ɛm* (Jer 50:12), *yonaqt'ô* (Job 8:16). In other noun forms, *ḥireq* or *segol* appears to be the regular reflex in unstressed syllables, although there are few examples. For *ḥireq* I have listed only *mêniqt'ô* (2 Kgs 11:2, 2 Chr 22:11), *gəbirt'î* (Gen 16:8, and similar cases, as Gen 16:9), also (presumably) *pəlišt'î* and related forms.

6.2 *Ṣere* occurs in stressed syllables of this sort:

(i) In word-initial position in the imperfect verb forms *way:'ebk* and *way:'ešt*, and also in the noun *q'edmɔ*, all three both in contextual and in pausal position. *Ṣere* also occurs in verb and noun forms of similar origin, in which the syllable has been secondarily opened, as *wat:'erɛʾ*, *s'efɛr*. *Segol* is common in such noun forms, but does not occur in verb forms, where the usual reflex of *i is *ḥireq* (see §6.1.i), even in some cases before the voiceless pharyngal *ḥet*, as *way:'iḥan*. *Pataḥ* may also occur before *ḥet*, as *way:'aḥaṣ*, before *he* as *wat:'ahar*, *ʿayin* as *way:'aʿal* and also, where segolation has not occurred, in *way:'aʾt* (Isa 41:25) and *way:'arʾ*.

(ii) *Ṣere* also occurs before a lengthened consonant which does not result from the assimilation of a root consonant to that of a pronominal affix, as *wənɔs'eb:ɔ* (1 Chr 13:3), *heḥ'el:û* (2 Chr 20:22). In other *hifʿil* perfect forms it occurs only where the (originally lengthened) final root consonant is not marked with *dagesh*, and so in forms from PRR, R^{cc}, but not in forms from SBB, MSS, QLL, where *pataḥ* occurs. *Ṣere* regularly occurs in in *hifʿil* imperfect forms (see BRR, HLL, ḤLL, PRR, ṢRR, R^{cc}, TLL).

(iii) *Ṣere* occurs in the stressed syllable of f.pl. forms of the imperfect or imperative *piʿel* in context, as *wəlam:'ednɔ* (Jer 9:19, also forms from DBR, RNN, ŚBR), also the *hifʿil* form *haʾz'en:ɔ* (Gen 4:23, Isa 32:9) and the *qal* form *tɔg'elnɔ* (Ps 48:12, 51:10, 97:8). Other *qal* forms show *pataḥ*, as *tel'aknɔ* (Ruth 1:11), as do all *nifʿal* forms, as *teʾɔk'alnɔ* (Jer 24:2). In *piʿel* and *hifʿil* forms, *pataḥ* occurs only before *ʿayin* (forms from BQ^{c} 2 Kgs 2:24, NB^{c} Ps 119:171), and in *tənɔʾ'apnɔ* (Hos 4:14), which is probably a case of "minor pause."[7]

(iv) *Ṣere* occurs exceptionally in closed stressed non-final syllables in pausal verb forms: f.pl. imperfect *piʿel tədab:'ernɔ* (Prov 24:2, with *silluq*) and (in minor pause) jussive forms *wəyag:'edkɔ* (Deut 32:7, with *zaqef*) and *təkab:'edkɔ* (Prov 4:8, with *revia mugrash*).[8]

---

[7] The form stands at the end of a clause at the main division of a long *atnaḥ* unit. Alternatively the occurrence of *pataḥ* may be due to the influence of the preceding *qameṣ* followed by *alef* (see §19.2).

[8] Both forms are examples of the use of a "short" imperfect form in a situation which is semantically (though not formally) the apodosis of a conditional sentence. See Driver 1892,

6.3 *Segol* occurs in a closed word-initial syllable under stress in *n'ɛgbɔ* and *q'ɛdšɔ* (in pause and in context, Gen 13:14, 28:14, Judg 4:9, 10). *Segol* also occurs in such syllables when unstressed, generally after *alef*, *ḥet* or *ʿayin* (See Revell 1985a, §3.1, 2, 3, 5). Where the syllable has been secondarily opened and is stressed, *segol* is the commonest reflex of *i in a word-initial syllable, both in context and in pause (see Revell 1985a, §§3.2, 3, 4, 5). Where the syllable is not word-initial, *segol* is the regular reflex of *i in contextual position (*pataḥ* occurs before gutturals). In pause it occurs only in *ʾom'ɛnɛt* (Ruth 4:16, see §6.4). In other forms, I have noted *segol* only in an unstressed closed syllable in *šəʾɛlt'ɛm* (1 Sam 12:13, 25:5, Job 21:29), *bəhɛmt'ô* (Gen 36:6 etc.), *miš:əkɛnt'ɔh* (Exod 3:22), *ḥăbɛrtək'ɔ* (Mal 2:14).

6.4 *Qameṣ* occurs as a reflex of *i in closed non-final syllables only under pausal stress. Among verb forms, *qameṣ* is the regular reflex in *qal* perfect forms, as *ʾɔh'ɔbtî*. In perfect *piʿel* and *hifʿil* forms *qameṣ* is regular only before *ḥet* and *ʿayin*. Elsewhere, *qameṣ* occurs mostly in forms in the three "poetical" books. *Pataḥ* occurs occasionally in pause in perfect *qal* forms, commonly in perfect forms from other stems (see §19). *Pataḥ* is the regular reflex in pause in imperfect verb forms (see §6.2.iii). Among noun forms, *qameṣ* occurs under pausal stress in secondarily opened syllables, sporadically in word-initial position, as *š'ɔbɛt* (see Revell 1985a, §§3.1, 2, 3, 5), and regularly where the syllable is not word-initial (f.s. *qal* participle forms from ʾKL, YŠB, NʾP, SRR, ʿMD, ŠMʿ, ŠQṬ, also *gəb'ɔrɛt* Isa 47:7, *šal:ʾɔṭɛt* Ezek 16:30, and presumably *pəl'ɔšɛt*, and *ḥid:'ɔqɛl* Dan 10:4). However *segol* occurs in the f.s. *qal* participle *ʾom'ɛnɛṯ* (Ruth 4:16), *pataḥ* in *bor'aḥat* (Gen 16:8).

7. *Syllables which were originally closed, and are closed, stressed, and final in MT.* In verb forms, syllables of this sort were originally either final (-VC), or penultimate (-VC:V > -VC). In nouns, only the latter type occurs. The reflex of *i in such syllables is generally *pataḥ* or *ṣere*. In nouns, and in verbs in pausal situations, *pataḥ* is a somewhat more common reflex of *i in this situation than is *ṣere*. *Ṣere* is considerably more common than *pataḥ* in verbs in contextual situations. Syllable structure and position clearly have little influence on the development of *i in this situation. The most important conditioning factor appears to be the phonetic environment of the vowel, as described in §12 and following.

8. *Syllables which were originally open, but are closed, stressed, and final in MT.* The reflex of *i in a syllable of this sort is generally *ṣere*, although *segol, pataḥ* and *qameṣ* may also occur. The following description deals with these latter cases in which *ṣere* does not occur.

8.1 *Segol* occurs occasionally in nouns which are frequently unstressed (as *bɛn*, Lev 24:10, see Yeivin 1980, §305). In verbs, *segol* occurs in *piʿel* perfect forms from KBS (see §14.2), DBR and KPR (see §14.4).

8.2 *Pataḥ* occurs (i) in verb forms:

(a) Before *ʿayin* and *ḥet*: *Pataḥ* always occurs in contextual situations, as *šɔm'aʿ*, *śɔm'aḥ*. *Ṣere* always occurs in pausal situations, including some "minor" situations before *ḥet*, as *ṣim:'eaḥ* (Ezek 16:7, at the end of a clause), or *tənat:'eaḥ* (Exod 29:17, at a major division within a clause).[9]

(b) Before other consonants: In perfect forms *pataḥ* and *ṣere* are about equally common, as *yɔš'en* and *dɔb'aq* in the *qal*, *ʾiz:'en* and *ʾib:'ad* in the *piʿel*. In imperfect forms, *pataḥ* occurs before *reš* in *nifʿal* forms from ʾMR and ŠBR, and in the *piʿel* form *wɔʾeḥ'ar* (Gen 32:5). *Ṣere* occurs in *nifʿal* forms from ZMR, NZR, NʿR, SGR, ʿBR, QBR, ŠBR, ŠMR, and in all other *piʿel* imperfect forms where the syllable was originally open and is stressed. Before consonants other than *reš*, *pataḥ* occurs in *qal* imperfect forms from roots Iʾ, as *yoʾk'al* (where *ṣere* also occurs, but is less common), and in *ter'ad* (Jer 13:17, Lam 3:48) from a root I*y*, where *ṣere* is usual. *Pataḥ* occcurs in a verb form in pause only in *qɔm'al* (Isa 33:9).

(ii) In noun forms, *pataḥ* occurs in substantives in the construct state (where *ṣere* also occurs, and is considerably more common, see §§21–25). *Pataḥ* also occurs in most *nifʿal* and some *piʿel* infinitive construct forms from roots III*ḥ* and IIIʿ which occur within clauses (see §22.1.iii).

---

§152. Since these "short" forms had no terminal vowel, the syllable has always been closed, unlike those discussed in §10. The shift of stress to the final vowel, which is typical of contextual forms, has not occurred because the forms stand in "minor" pausal situations: at the end of a stich of poetry, or at the main division of such a stich (see Revell 1985b, §§14, 18).

---

[9] The case in Exod 29:17 occurs at the main division of a short *atnaḥ* unit, noted as a minor pausal situation in Revell 1985b, §§14.5, 18.4.

8.3 *Qameṣ* occurs as a reflex of *i in syllables of this sort only in the pausal form of the 3m.s. perfect *qal* of Š˒L (1 Chr 4:10, and, in minor pause, Jos 19:50, Judg 8:26).[10] In other perfect *qal* forms in pause, the reflex of *i is *ṣere*, as *dɔb'eq* (2 Kgs 3:3), save for the use of *pataḥ* in *qɔm'al* (Isa 33:9). In verb forms other than perfect *qal*, the reflex of *i is always *ṣere* in pause, with the possible exception of *to˒k'al* (1 Sam 1:7).[11] *Ṣere* is also the regular reflex of *i in syllables of this sort in the absolute forms of nouns.

9. *Closed final syllables (of any origin) which are unstressed in MT.* The syllables considered here include those unstressed through retraction of the accent (*nesiga*), or through the use of *maqqef* (showing that the word is not stressed), as well as those regularly unstressed in contextual position in MT (mostly in imperfect verb forms with *waw* consecutive). The reflex of *i in syllables of this sort is generally *segol*, but *ḥireq*, *ṣere*, and *pataḥ* may also occur.

9.1 *Ḥireq* occurs in one verb form in context, *wat:'ɔriṣ* (Judg 9:53).

9.2 *Ṣere* occurs in most participles in which the final syllable is unstressed through *nesiga*, as *š'olep ḥ'ɔrɛb* (Judg 8:10), and occasionally in other forms in the same situation, as *g'ɔreš leṣ* (Prov 22:10), *wəyin:'ɔten lɔk* (Esth 5:3, 6, 7:2, 9:12, but not in the similar case in Lev 24:20). *Ṣere* also occurs occasionally in a word with *maqqef*, as *dab:er* (Exod 20:19, Ezek 14:4), *šen* (1 Sam 14:4). Once it occurs in a verb form in context, *way:'ɔṣeṣ* (Num 17:23).

9.3 *Pataḥ* occurs (i) most commonly in verb forms, as follows:

(a) Before *ḥet* or *ˁayin*, as *way:'ɔraˁ*.

(b) Before *reš* in *waw* consecutive imperfect *hifˁil* forms from SWR, ṢRR, and ŠYR, but not from ˒WR or PRR.

(c) In other *waw* consecutive imperfect verb forms from ˒KL (*qal*), and ˁWD (*hifˁil*), but not in similar *qal* forms from ˒ḤZ, ˒MR, ˒SP, or elsewhere after *ˁayin*.

(d) Elsewhere in verb forms only where the syllable is unstressed with *maqqef* in *nit:an* (Judg 16:5), or through *nesiga* in *teˁ'ɔzab* (Job 18:4).

(ii) In noun forms, the reflex of *i in an unstressed syllable closed by *ḥet* or *ˁayin* is regularly *pataḥ*. Elsewhere, *pataḥ* occurs in syllables unstressed through the use of *maqqef* in those nouns of form *CiC: which also show *pataḥ* in the absolute singular forms, as *bat*, and also in the construct form *qan* (Deut 22:6). *Pataḥ* also occurs sporadically in participles as listed in §23.1, and in a *piˁel* infinitive construct form from ḤRḤR (Prov 26:21).

10. *Base-final syllables before a 2m.s. or 2p.pl. suffix.* These syllables are unstressed. They were originally open, but appear usually to have become closed in the reading tradition represented in MT.[12] This is suggested by the fact that *segol* is the common reflex of *i in such syllables. *Ḥireq*, *ṣere* and *pataḥ* also appear, but, where *ṣere* is used, the syllable was probably open. The following paragraphs list the cases in which *segol* is not used.

10.1 *Ḥireq* occurs (i) in word-initial position in forms from *ben*, *šem*, and in *hăyišk'ɛm* (Deut 13:4). *Segol* occurs in four other forms from *yeš*, each preceded by *˒im* with *maqqef*.

(ii) In *qal* participles from ˒SP and ˒YB, and in *piˁel* forms from ˒MṢ, ˒SP, PRŠ, QDŠ, as *˒osipk'ɔ* (2 Kgs 22:20), *˒ă˒am:iṣk'ɛm* Job 16:5. *Segol* occurs in other *qal* participles, and in most other *piˁel* forms (but see §§10.2, 3), including forms from ˒LP, ḤLṢ, KZB, ŚGB, ŠRŠ, in which the vowel is similarly followed by *p* or *b*, or a sibilant.

10.2 *Ṣere* occurs in the *piˁel* of ŠLḤ, in an infinitive form, *bəšal:eḥăk'ɔ* (Deut 15:18), a participle (Jer 28:16), and in imperfect forms (Gen 26:29, 31:27, 32:27), but not in perfect forms (see §10.3). *Ṣere* also occurs in a *qal* participle form from ŠLḤ in 1 Sam 21:3. In all these cases the *ḥet* has *ḥaṭef pataḥ*, so the syllable would appear to have been open (but see §10.3).

---

[10] Note that in perfect *qal* forms from this root the reflex of *i is *pataḥ* or *qameṣ* where the *alef* is preceded by *qameṣ*, as *wəšɔ˒alt'ɔ* (Deut 13:15), and, in pause, *šə˒'ɔl*, *šɔ˒'ɔlɔ* (cf. *˒ɔh'eb*, *ṭɔh'erɔ* Lev 12:8), but not where the preceding vowel is *shewa*, as *ûšə˒ɛlt'ɛm*, and, where the syllable is open, *ûšə˒el'ek* etc. (See also §19.2).

[11] If this form was originally a "preterite," the final syllable would have been originally closed, so that *pataḥ* is an expected reflex of *i (see §7). However the "long" form of the preceding *wat:ibk'ɛ* (where the short "preterite" form is expected) shows that the origin of the vowelling here is questionable.

[12] The fact that the *kaf* of the pronoun has no *dagesh* reflects the fact that the preceding syllable was once open, so that the fricative realization of the consonant developed. This development was not, of course, reversed when the preceding syllable subsequently became closed.

10.3 *Paṭaḥ* occurs in perfect *piᶜel* forms: in *šil:aḥăk'ɔ* (1 Sam 20:22, despite the *ḥaṭef paṭaḥ* under the *ḥet*), and in nine cases from BRK. *Segol* occurs in a perfect *piᶜel* form only in *wəšerɛšk'ɔ* (Ps 52:7).

11. *Syllable Structure and Position: Conclusions.* It is clear from this description that the change from *i to *paṭaḥ* takes place most readily in an originally closed syllable, particularly if it is non-final. This is already well-known, but the significance of two additional considerations which are not generally accepted is also brought out by this survey.

11.1 Stress is probably not a particularly important factor. The failure of *paṭaḥ* to develop in unstressed syllables is generally to be accounted for by the fact that the syllable was originally open (§§8, 10), word-initial (§§6.1.i, 6.2.i, 6.3.i), or word final (§9, *paṭaḥ* often fails to occur in word final syllables even where originally closed and under stress, §7), or by the fact that the vowel is followed by a lengthened consonant (§§6.1.ii, 6.2.ii). In closed non-final syllables which are unstressed, *paṭaḥ* occurs regularly in verb forms and in *qal* participles, but not in other noun forms (§6.1.iii). It could be argued that the use of *paṭaḥ* in *qal* participles is due to the influence of original long *a in the preceding open syllable (see §19.2). However there is no obvious reason for the occasional failure of *paṭaḥ* to appear in verb forms (see §§6.1.ii, 6.1.iii, 6.3), and the vowelling of participles differs from that of other nouns in other ways (see §23). The balance of probability is, then, that stress does not determine the reflex of *i in these syllables, although it certainly is a conditioning factor.

11.2 Both *ṣere* and *paṭaḥ* appear as possible reflexes of *i in every one of the situations discussed. Clearly both vowels are normal results of a natural process of vowel change; it is unreasonable to argue that, in one group of these situations, *ṣere* is a surprising disruption of the normal pattern of change, explicable only by appeal to analogy. Either the two vowels are free variants, or other conditioning factors must be involved.

PHONETIC ENVIRONMENT

12. Where *i gives rise to more than one MT reflex in syllables of the same structure, and in the same position relative to the boundaries of the word and to its stress pattern, it appears that the reflex of the vowel is often determined by its phonetic environment. This can be clearly demonstrated in those forms in which either *paṭaḥ* (*segol*) or *ṣere* appears in a significant proportion of the cases in any given situation.[13] The following forms are used for this demonstration.

NOUN FORMS: absolute and construct.
- *CiC:, as *bat, leb.*
- *CiCC, as *ṣ'ɛdɛq, s'efɛr.*

NOUN FORMS: construct forms only.
- *CaCiC, as *ḥăṣ'ar ᶜăq'eb.* Absolute forms of these nouns regularly show *ṣere*. Construct forms of other substantival nouns which are not monosyllables regularly show *paṭaḥ* (see §22). Absolute forms of such nouns, and also all forms of participles, regularly show *ṣere*.

VERB FORMS: perfect masculine singular in contextual position.
- *qatil (*qal*), as *dɔb'aq, yɔ̆š'en.*
- *qat:il (*piᶜel*), as *ʾib:'ad, ʾiz:'en.*
- *haqit: (*hifᶜil* from geminate roots), as *hed'aq, hes'eb.*

VERB FORMS: imperfect with no pronominal affix (3m/f.s., 2m.s., 1c.s/pl.) in contextual or pausal position.
- *qal* forms of the patterns *yoʾk'al, yoʾḥ'ez* and *yed'aᶜ, yeš'eb.*

*The Consonant Following *i*

13. Where *i is followed by *ḥet* or *ᶜayin* in these forms, the MT reflex is regularly *paṭaḥ*.[14] Other consonants following *i do condition its development to some extent, as shown in Rabin 1968. However the consonant preceding *i generally has a more significant effect on its development than that following it. Consequently the following description is based on the consonant preceding the vowel.

*The Consonant Preceding *i*

14. *Stops.* Where *i is preceded by a stop, the environment is clearly conducive to the development of

[13] A preliminary demonstration is given in Revell 1985a, §4. In the following description, an environment giving rise to MT *segol* is considered to be more favorable to the change to *paṭaḥ* than one giving rise to MT *ṣere*, as justified by §5 above, and by Revell 1985a, §5.

[14] The only exception in the forms as listed is the use of *qameṣ* in the imperfect of YDᶜ in pause. However *ṣere* occurs in some forms not included in this section, as in nouns of form *CaCiC in the absolute state, as *yɔp'eaḥ*, or verbs in pausal situations, as *biq:'eaᶜ* (2 Kgs 15:16).

pataḥ, although more so where the stop is voiced (*b*, *d*, *g*) or emphatic (*ṭ*, *q*) than where it is voiceless (*p*, *t*). *Pataḥ* occurs in the following forms:

NOUN FORMS
*CiC:, *bat, gat, qan* (construct only), *pat*.
*CiCC, *segol* is the usual reflex of *i (see Revell 1985a, §§3.3, 4, 5).
*CaCiC (construct): *zəq'an, ḥăd'al, kəb'ad, šək'an, yət'ad*.
VERB FORMS
*qatil, GBR, DBQ, KBD, LBŠ, RBṢ, GDL, ḤDL, QDŠ, ŠKB, ŠKN (in context).
*qat:il, ʾBD, GDL, ZQQ, ḤBR, KTT, KLKL, MDD, NBṬ, NGŠ, NKR, NQP, NQR, SGR, ʿPR, QDŠ, ŠBR, ŠQṢ, also, with *segol*, DBR, KPR, and, in some cases, KBS (see §14.2).
*haqit:, DQQ, GLL, TLL, (also PRR and TZZ in pause).
Imperfect *yoʾb'ad* (contextual), *yoʾk'al* (contextual, and *waw* consecutive forms in pause).

14.1 *Ṣere* is the regular reflex where *i stands between a stop and *nun*.[15] Thus:

NOUN FORMS
*CiC:, *ken* (both "base" and "gnat"), *qen* (absolute).
VERB FORMS
*qatil, ZQN
*qat:il, MGN, NGN, ŠKN (in pause, including "minor" positions in Deut 33:20, Judg 5:17), TKN, TQN.

Pataḥ occurs (as indicated above) in *CiC: *qan* (unstressed in construct, Deut 22:6), in *CaCiC (construct) *zəq'an, šək'an*, and in *qatil *sɔk'an* in context. It is striking that in all these cases in which *i followed by *nun* gives rise to *ṣere*, the preceding stop is a palatal. However this may be a coincidence, as the only case in which *ṣere* fails to appear where *i was preceded by some other stop appears to be the *CiCC form *ṭ'ɛnɛʾ*.

14.2 *Ṣere* is a common reflex where the vowel is followed by a sibilant. Thus:

NOUN FORMS
*CiC:, *gez, qeṣ*.
*CiCC, *g'ezaʿ, g'ezɛl* (both construct) *k'esɛl* (before a modifier Ps 49:14; *segol* occurs at the end of a phrase in Qoh 7:25), *p'ešɛr* (construct), *t'ešaʿ*. However *segol* occurs in *b'ɛṣaʿ, g'ɛšɛm, g'ɛšɛt, d'ɛšɛ', d'ɛšɛn, p'ɛṣaʿ, p'ɛšaʿ, q'ɛṣɛp, q'ɛšɛr*.
VERB FORMS
*qatil, *yɔb'eš, ḥɔp'eṣ*. *Pataḥ* occurs in *lɔb'aš, qɔd'aš*.
*qat:il, *biq:'eš, nip:'eṣ, nit:'eṣ*, also *kib:'es* within a clause in Gen 49:11, 2 Sam 19:25; *kib:'ɛs*, with *segol*, occurs elsewhere.[16] *Pataḥ* occurs in *nig:'aš, qid:'aš, šiq:'aṣ*.

14.3 *Ṣere* also occurs after voiced or intensive stops in the construct noun *ʿăq'eb* (Gen 25:26), and in the imperfect *qal* form *yoʾb'ed* (pause only). After voiceless stops, *ṣere* occurs in *CiC: *tel* (also with *maqqef* Jos 8:28), *ṭeṭ* (*segol* in *lɔṭɛṭ* with *maqqef*), *CiCC *p'etaḥ* (Ps 119:130), *t'eqaʿ* (Ps 150:3, both construct), *qat:il *šik:'el*, *haqit: *hep'er* (contextual, *pataḥ* in pause). In the *qal* imperfect of ʾKL, *ṣere* occurs in pause only, including the 1c.s. imperfect form with *waw* consecutive. *Pataḥ* occurs in all forms in context, and in pause in *waw* consecutive imperfect forms other than 1c.s.

14.4 The expected change to *pataḥ* is not completed in *i where it stands between a labial and *reš* in perfect *piʿel* forms of DBR and KPR. *Segol* occurs in contextual position, *ṣere* in pause (DBR only, including "minor" positions as Jos 14:10). However *pataḥ* does occur in this situation in forms from ḤBR, ʿPR, ŠBR.

15. *Sibilants.*

15.1 Where *i is preceded by a voiced or intensive sibilant (*z*, *ṣ*), the environment is also conducive to the development of *pataḥ*. *Pataḥ* occurs in *CiC: *ṣad*, *segol* in *CiCC *z'ɛbaḥ, ṣ'ɛdɛq, ṣ'ɛmɛd, ṣ'ɛmaḥ*, *pataḥ* in *CaCiC (construct) *ḥăṣ'ar*, *qatil *hɔz'aq*, *qat:il *biz:'ar, hiz:'aq, piz:'ar, qiṣ:'aṣ* (within a clause, 2 Kgs 18:16) *qiṣ:'ar*, *haqit: *heṣ'ar*. *Ṣere* occurs only in *CiC: *ṣel*, *CiCC *z'ekɛr*, and *qat:il *qiṣ:'eṣ* (Ps 129:4, at the main division of a stich of poetry, a minor pausal position, see Revell, 1985b, §§14.4, 18.3).

15.2 Where *i is preceded by a voiceless sibilant and is followed by *nun* or a voiceless consonant, the environment is clearly not conducive to the development of

[15] Cf. also the noun *mɔg'en* (with lengthened final consonant), and the imperfect *qal* of NTN, which shows *ṣere* in the final syllable save before *maqqef*, where *segol* is usual, but *pataḥ* occurs in Jud 16:5.

[16] *Segol* occurs in *kib:'ɛs* where the form has prefixed *waw* consecutive, and is followed by *begɔd'ɔw*, either immediately (nine cases) or with one or two words intervening (three cases). *Ṣere* occurs where neither feature is present. The situations in which the two forms are used thus show a clear semantic difference, but the relation between this and the difference in vowelling can be a matter of speculation only.

paṭaḥ. *Ṣere* occurs in this situation in *CiC: *šen*, *šeš*, CiCC *s'epɛl*, *s'epɛr*, *s'eṭɛr*, *š'epaḥ*, *š'epɛl*, *š'ešɛṭ*, and in *qatil *yɔš'en*. *Segol* occurs in *CiC: *šɛn* where closely joined to what follows, and in *CiCC *š'ɛkɛm*. Both *ṣere* and *segol* occur in *CiCC *š'ekɛl* (see Revell 1985a, §6.4). *Paṭaḥ* occurs only in *CiC: *sap*, *qatil *ʿɔš'an*.

15.3 Where *i is preceded by a voiceless sibilant, and followed by a voiced labial (*b*, *m*), the environment is somewhat more conducive to the development of *paṭaḥ*.[17] *Segol* occurs in *CiCC *š'ɛbaʿ*, *š'ɛbɛṭ*, *š'ɛmɛš*, and in most forms of *š'ɛbɛr*. *Paṭaḥ* occurs in *qatil *ʾɔš'am* (within a clause Lev 5:19, Num 5:7), *qat:il *ḥiš:'ab*. *Ṣere* occurs in *CiCC *s'ebɛl*, *š'ebeṭ*, *š'emaʿ*, *š'emɛṣ*, and also in some construct forms of *š'ebɛr*, in *qatil *ʾɔš'em* (at the end of the clause Lev 5:17, 23, also within a clause in Lev 5:4, where it is the second of two closely related verbs, a common position for "minor" pausal forms, see Revell 1985b, §§11.1, 14.6, 18.5). *Ṣere* also occurs in *haqit: *hes'eb*, and the imperfect *qal* form *yeš'eb*, including a *waw* consecutive form in pause in Ruth 4:1.

15.4 Where *i is preceded by a voiceless sibilant and followed by other consonants, the environment is generally conducive to the development of *paṭaḥ*. *Segol* occurs in *CiCC *š'ɛqɛl*, *š'ɛqɛr*, *š'ɛlah*, also in *š'ɛṭɛp* absolute, including Prov 27:4, cf. LXX, Targum. *Paṭaḥ* occurs in *CaCiC (construct) *ḥăs'ar*, *qatil *yɔš'ar*, *qat:il *biš:'ar*, *ḥiš:'aq*, *yis:'ad*, *yis:'ar*. *Ṣere* occurs only in *CiCC *š'eṭɛp* in the construct (Ps 32:6).

16. *Gutturals*. Where *i is preceded by a laryngal or pharyngal the environment is not conducive to the development of *paṭaḥ*. The MT reflexes occur as follows:

NOUN FORMS

*CiC:. *Paṭaḥ* occurs in *ʾaṭ* (also presumably in the contextual form corresponding to the pausal *hɔṭ* "fear," Job 41:25). *Ṣere* occurs in *ʾem*, *ʾeb*, *ʾes*, *ʾeṭ* (both noun and preposition) *ḥek*, *ḥen*, *ḥeṣ*, *ḥeṭ*, *ʿez*, *ʿeṭ*.

*CiCC. *Segol* occurs in four nouns in which the vowel is preceded by *he* or *ḥeṭ*. *Ṣere* occurs in 26 nouns in which the vowel is preceded by these consonants (see Revell 1985a, n. 12, 18) and in all nouns in which the vowel is preceded by *alef* or *ʿayin* (See Revell 1985a, n. 11).

VERB FORMS

*qatil. *Paṭaḥ* occurs in *šɔʾ'al*, and in *ʾɔh'ab* in four cases in contextual position. *Ṣere* occurs in *ṭɔh'er*, *rɔʿ'eb*, and in *ʾɔh'eb* in pause, and also in Gen 27:14, where the form is clearly not in a pausal position.

*qat:il. In this form, in which the consonant was originally lengthened, *paṭaḥ* occurs commonly after *he* and *ḥeṭ* (ʾHR, THR, LHṬ, MHR, NHG, NHL, NḤM, NHT, RḤM, RḤQ). However *ṣere* occurs in KHN, KHD, KHŠ, ŠHT, and always after *alef* (BʾR, MʾN, NʾṢ, NʾR) and *ʿayin* (BʿR, NʿR).

17. *Sonants*. This title is applicable to the remaining consonants (*w*, *y*, *l*, *m*, *n*, *r*); however they form a diverse group, and so must be treated separately. There is little information on some of them, and virtually none on *waw*, which is therefore not included.

17.1 *Reš* provides an environment conducive to the change of following *i to *paṭaḥ*.[18] In nouns of form *CiCC, *segol* occurs in *r'ɛbaʿ*, *r'ɛdɛṭ*, *r'ɛkɛb*, *r'ɛsɛn*, *r'ɛšaʿ*, *r'ɛšɛp*, *r'ɛšɛṭ*, *ṣere* only in *r'ebɛṣ* (construct). In *CaCiC (construct) *paṭaḥ* occurs in *ʿăr'al*. In verb forms, *paṭaḥ* occurs in *qatil *qɔr'ab*, *qat:il *ʾer'aš*, *per'aš* and in *ber'ak* where it is followed by a modifier. *Ṣere* occurs in *ṭer'ep*, *šer'eṭ*, and in *ber'ek* where it is not followed by a modifier (Num 23:20, Ps 10:3). In the imperfect *qal* of YRD, *paṭaḥ* occurs in *waw* consecutive forms in pause, and in other forms in Jer 13:17, Lam 3:48. *Ṣere* occurs elsewhere.

17.2 *Mem* provides an environment conducive to the change of *i to *paṭaḥ*, except where the vowel is followed by a sibilant.[19] In this latter situation, *ṣere* occurs in *CiC: *mes*, *CiCC *m'ezah*, *m'eṣah*, *qat:il *ʾim:ʾeṣ*, *ḥim:'eṣ*. *Paṭaḥ* occurs only in the *CiC: form *mas*.[20] Before other consonants, *paṭaḥ* occurs in *CiC: *mad*, *segol* in *CiCC *m'ɛkɛr*, *m'ɛkɛs*, *m'ɛṭɛg*, *paṭaḥ* in *qatil *qɔm'al* (in pause Isa 33:9), *qat:il *lim:'ad*, *haqit: *hem'ar*, and in the imperfect *qal* of ʾMR in context. *Ṣere* occurs only in the imperfect *qal* of ʾMR in pause.

---

[17] See *mes'ab* and *mes'eb* (with lengthened final consonant) described in §24.

[18] Cf. also the noun *mor'ag* (with lengthened final consonant).

[19] Cf. also the *ṣere* in the noun *hɔm'eš* with *segol* in *ʾɛm'ɛṭ* (both with originally lengthened final consonant). Cases of *šɔm'en* with *ṣere* are taken as nominal, not as *qal* perfects. However *nun* following *i might well have inhibited the change to *paṭaḥ*. See §18.1.

[20] The original vowel of this form is, in fact, uncertain. See Blau 1981, §4.1.1.

17.3 *Lamed* provides an environment which is somewhat conducive to the change of following *i to *patah*, but not strongly so. In nouns of forms *CiC:, *ṣere* occurs in *leb* unless it is closely joined to what follows, in which case *segol* occurs. In nouns of form *CiCC, *segol* occurs (*l'ɛdɛt*, *l'ɛkɛt*, *l'ɛqah*). In *qit:al, *patah* occurs where the vowel is followed by a stop (*mil:'at*, *pil:'ag*), *ṣere* elsewhere (*hil:'el*, *ḥil:'el*, *hil:'eṣ*, *mil:'el*, *qil:'el*). *Patah* occurs in the *qal* imperfect of HLK in pause in *waw* consecutive forms (also a jussive form Job 27:21). *Sere* occurs in other forms, and also in the imperfect *qal* of YLD.

17.4 *Yod* provides an environment which appears to be somewhat conducive to the change of following *i to *patah*, but the information is scanty. It comes almost only from nouns of form *CiCC. In these forms, the reflex of *i is *segol* where the vowel is followed by a stop (*y'ɛqɛb*, *y'ɛtɛr*); *ṣere* where it is followed by a sibilant (*y'eṣer*, *y'ešɛr*), save that both vowels are used in *y'eša*ᶜ.[21] The only other information is provided by the *qat:il form *qiy:'am*, showing *patah*.

17.5 *Nun* provides an environment which is not conducive to the change of following *i to *patah*. The development of the vowel may be influenced to some extent by the following consonant. The information is provided almost solely by nouns of form *CiCC. In these forms, *segol* occurs before palatal stops (*n'ɛgɛb*, *n'ɛgɛd*, *n'ɛga*ᶜ, *n'ɛkɛd*) and sometimes before other voiced stops (*n'ɛbɛl*, *n'ɛdɛr*). *Ṣere* also occurs in these last two forms, and before other stops (*n'eṭɛl*, *n'epɛl*, *n'etah*). Before sibilants, *segol* occurs in *n'ɛzɛm*, *n'ɛšɛk*, *n'ɛšɛp*, *n'ɛšɛr*; *ṣere* in *n'ezɛq*, *n'ezɛr*, *n'eṣɛr*. Both vowels occur in *n'esɛk*, *n'esah*, *n'ešɛq*. The only other information is provided by the *CiC: form *nes*, showing *ṣere*.

18.1 *Phonetic Environment: Conclusions*. It is clear from the survey in §§14–17 that the development of *i is conditioned, in many cases, by the preceding consonant. *Patah* is most likely to develop after a consonant which is voiced (other than ᶜ, *y*, *l*, *n*), and/or a stop. The development of *i in a closed syllable is also conditioned by the following consonant. The following sound is only markedly conducive to the development of *patah* where it is a pharyngal (§13). However there is some evidence that a palatal stop following the vowel may also be conducive to the development of *patah* (§§17.3, 4, 5), so also possibly dental stops (§§17.3, 4). A following *nun* definitely inhibits the development of *patah* (§§14.1, 15.2), as does a sibilant (§§14.2, 17.2, 4), and, in some cases, a labial (§15.3) or other voiceless consonants (§15.2).

18.2 The influence of the phonetic environment is no doubt always one of the factors conditioning the development of *i, although its influence may be less marked in other cases than in those used for illustration above due to the greater influence of other factors. The *pi*ᶜ*el* forms with 2m.s. or 2p.pl. pronominal suffix described in §10 provide an example. In *pi*ᶜ*el* perfect forms, the reflex of *i in the syllable before the suffix is *patah* before *ḥ* and between *r* and *k*, that is, in environments favorable to change. The reflex is *segol* after *r* and before a sibilant, an environment unfavorable to change. In other *pi*ᶜ*el* forms, the common reflex of *i in this situation is *segol*, but *ḥireq* appears where the environment is unfavorable to change. Thus *ḥireq* occurs before *p* after a sibilant in ᵓSP,[22] but not after *l* in ᵓLP. *Ḥireq* also occurs before a sibilant in PRŠ, QDŠ, including the emphatic *ṣ* in ᵓMṢ (cf. §17.2), although not in ḤLṢ.

19.1 *Phonetic Environment and the Development of* Qames. The development of *qameṣ* as a reflex of *i in the closed stressed non-final syllables of perfect verb forms in pause also seems to be conditioned, to some extent, by the phonetic environment.

(i) In such syllables, the reflex of *i is regularly *qameṣ* where the vowel is followed by *ḥet* or ᶜ*ayin*. The only exception is *šiw:'a*ᶜ*tî* (Ps 88:14), one of a number of cases in which pausal vowelling does not occur in a word marked with *atnah* acting as the main divider of a verse in the Three Books.[23]

---

[21] *Sere* occurs where *y'eša*ᶜ stands within a clause, in construct, or otherwise (Hab 3:13, 13; Ps 12:6, 20:7, 50:23). *Segol* is used where the word is in pause (Ps 132:16; Job 5:4, 11) or stands at the end of a clause elsewhere (Isa 45:8, 61:10).

[22] So also in the *qal* participle of ᵓSP (see §10.1.ii). *Hireq* also occurs in ᵓ*oyibk'ɔ* (Exod 23:4 and six other cases), the *qal* participle of ᵓYB, where the *b* was very likely voiceless before *k*. Cf. the warning in Talmud Yerushalmi on Mishna Berakot 2:3 (ed. Venice 1523, fol. 4, verso col. 2) that *tizkərû* (Num 15:40) must be pronounced carefully. According to the commentators, this is to avoid the pronunciation *tiskerû*. This suggests that devoicing of a fricative before a voiceless stop, the source of this unintentional blasphemy, was a common phenomenon. Similar devoicing through partial assimilation was also common in Syriac, as described in Jacob of Edessa's grammar (c. 700 A.D. See Merx 1889, Syriac p. 78).

[23] As with segolate forms such as ᵓ*'ɛreṣ* (Prov 30:21), and others in Ps 35:19, 48:11, 120:2.

(ii) In such syllables, the reflex of *i followed by *n* is *qameṣ* only in a *hifʿil* forms from ᵓMN in Ps 119:66. *Pataḥ* occurs in forms from the *qal* of ZQN (Gen 18:13, 27:2, Ps 37:25), and from the *hifʿil* of ŠKN (Ps 139:3). Preceding *m* is generally favorable to the change to *pataḥ* (see §17.2), a preceding palatal and following *n* provides a context unfavorable to change (see §14.1).

(iii) Elsewhere, *qameṣ* appears in such syllables in *qal* forms except in *wɔm'at:ī* (Gen 19:19). The following lengthened consonant is unfavorable to change (see §§6.1.ii, 6.2.ii, and cf. *m'ɔtnū*, with *qameṣ*, in Num 14.2, et al.).

(iv) In stems other than *qal*, *pataḥ* is the usual reflex of *i in the Twenty-one Books, appearing in *piʿel* forms from DBR, ḤRP, ŠBR, and in *hifʿil* forms from ḤRM, NGD, NSL, ŠBT. *Qameṣ* appears only in a *piʿel* form from ḤLL in Ezek 22:8, and in a *hifʿil* form from QDŠ in 2 Chron 29:19. In the Three Books, *qameṣ* is significantly more common, appearing in *piʿel* forms from HLK, YHL, YŠR, and in *hifʿil* forms from YHL, and QDŠ. *Pataḥ* occurs in *piʿel* forms from MGR, SPR, and in *hifʿil* forms from ᵓBD, ŠKL.

19.2 *Qameṣ* thus regularly appears in these closed stressed non-final syllables in *qal* forms in pause, but much less commonly in forms from other stems. This may indicate that the presence of *a in an open syllable was conducive to the change of *i in the following syllable to *pataḥ*. The fact that *pataḥ* regularly appears as the reflex of *i in the closed stressed non-final syllable of f.pl. imperfect forms in the *nifʿal*, but not in the *piʿel* (see §6.2.iii) may reflect the same conditioning factor. The influence of *a (MT *qameṣ*) in the preceding open syllable may be a partial cause of the appearance of *pataḥ* as reflex of *i in closed non-final unstressed syllables in *qal* perfect forms. Cf. its failure to appear where the MT vowel of the preceding syllable is not *qameṣ* in forms from Šᵓl and YRŠ (see §§6.1.iii, 6.3). In a similar way, preceding *a in an open syllable may be the cause of the development of *pataḥ* in the closed stressed final syllable of the *qatil forms from ᵓHB and Šᵓl (anomalous, see §16). A similar influence may have caused the appearance of *pataḥ* in the final syllable of *qal* imperfect forms of the *yoᵓk'al* type, where it is common in comparison to forms of the *yeš'eb* type (see §28.1). The vowel here was *a, changing to long *ā* and then to *ḥolem*. The presence of original long *ā (MT *ḥolem*) may have been the cause of the development of *pataḥ* in the closed unstressed non-final syllables of f.s. *qal* participle forms with suffixes (see §6.1.iii). Altogether, there is a considerable amount of evidence to suggest that the development of *i was affected by some reflexes of *a or *ā in a preceding open syllable.[24] However other factors may also have been at work in some of the cases mentioned here (see §28.2).

19.3 It appears, then, that *qameṣ* generally developed as a reflex of *i in environments favorable to the change of *i to *pataḥ*, and fails to develop in environments unfavorable to that change. The development of *qameṣ* must, then, be mainly due to the fact that, in such environments, the change from *i to *a* was quickly completed. The resulting *a* then changed to *qameṣ*, as did original *a, before the sounds were recorded in writing. In unfavorable situations, the change from *i to *a* took place much more slowly, so that the completion of the change to *pataḥ* required the whole period available, or, in many cases, was not completed within the period. The change to *qameṣ* did not take place in all situations favorable to the change to *pataḥ*. The development of *qameṣ* is favored by conditions different from those required for the development of *pataḥ*.[25] The fact that *qameṣ* occurs more frequently as a reflex of *i in the Three Books than in the Twenty-one presumably shows that conditions for the development of *qameṣ* were more favorable in the one corpus than in the other, possibly due to the use of different forms of chant, as suggested by the accentuation of MT.[26]

RELATION TO CONTEXT

20. In the above description, it has several times been noted that the MT reflex of *i is conditioned by the relationship of the word to its context, using the traditional terminology "absolute/construct" and "pausal/contextual." These terms refer to pairs of forms distinguished by differences in vowel pattern and/or stress position. These differences are due to ordinary

[24] In other probable cases of vowel harmony, the vowel seems to be affected by that of the following syllable, as in the masoretic descriptions of the sound of *shewa* before a guttural (see n. 5). Partial harmonizing (due to the lip position required for the following vowel?) appears to occur in the reflex of *a in the pronominal suffixes *-'anī* and *-'ɔnū*, and (if the argument in §32 is correct) before a historically lengthened guttural followed by *qameṣ*.

[25] *Qameṣ* develops more readily in an open syllable than in a closed one, and its development appears to be more strongly influenced by the following consonant than by the preceding. See Revell 1981, 84–86.

[26] Other features also differ in the two bodies of material. *Nesiga* is significantly more common in the Three Books than in the Twenty-one, in proportion to the number of possible cases. See Revell 1987a, §1.17.

processes of sound change which are general in the language, and not specific to either form in either pair. Forms showing the variations with these names, and some showing other variations, can be assigned to a single category of contrasting forms which may be called "terminal" and "medial."[27] Such forms are terminal or medial in the units or sub-units of speech into which the text was divided in the form of the reading tradition reflected by the vowel and stress patterns of MT. The extent to which any given form may be used as terminal or medial—that is, whether it can occur as a terminal form only at the end of speech units, or also at the end of sub-units, or whether it is restricted to medial position within sub-units, or may occur anywhere within a unit—depends on the structure and function of the form in question.[28] Terminal forms in MT are most commonly marked by stress on the penultimate syllable where the final is open, as *p'ɛrî, y'ɛhî, yip:ɔr'edû*, or by *qames* as reflex of *a in the stressed syllable, as *dɔb'ɔr, ʾɔk'ɔl*, or by both, as *ʾ'ɔt:ɔ, wɔḥ'ɔt:û*. Medial forms are most commonly marked by stress on a final vowel, as *pər'î, yəh'i, yip:ɔrəd'û*, or by *patah* as reflex of *a in a closed stressed syllable, as *dəb'ar, ʾɔk'al.*[29] The factors which condition the reflex of *a in forms such as those listed above, also condition the reflex of *i in similar circumstances.

21. *Terminal Position.* It has already been suggested that stress in pausal position (a form of terminal position) has greater effect on vowel change than does stress in contextual position (a form of medial position). This is demonstrated (among other things) by the fact that *qameṣ* as a reflex of *i appears only in pausal position (§§6.4, 8.3). Thus the maximum extent of vowel change was attained where the form was terminal in a speech unit, that is, where it was totally separated in speech from what followed. Where the syllable was word-final in MT, the development of pausal vowelling was conditioned by the original structure of the syllable, so that, where the syllable was originally open, the common reflex of *i is *sere* (*patah* is exceptional). Where the syllable was originally closed, the common reflex of *i is *patah*, although *ṣere* is not infrequent. The development of the vowel in contextual position was not so conditioned, so that the reflex of *i is commonly *ṣere*, less commonly *patah*, in syllables of both types. This suggests that the change of *i towards *a* began earlier in terminal position than in medial position, so that, by the time the change began to affect syllables in medial position, originally closed final syllables no longer differed from originally open ones. It is presumably the greater length of time over which the change was in progress which has brought about the greater degree of change in stressed syllables in pausal position, and not any significant difference between stress there and that in contextual position.

22. *Medial Position: Originally Open Syllables. Patah* occurs as a reflex of *i not only in originally closed final syllables under pausal stress, but also in originally open final syllables under contextual stress in construct nouns and in verb forms followed by subject or modifier. That is, it occurs in a syllable which, although final in the word in isolation, is not final in the syntactic unit which it forms. In this non-final position, the vowel undergoes the same change as it does in a stressed syllable which is internal in a word. The retention of the original *t* of the feminine gender marker **-at* in construct nouns is a similar phenomenon. A construct noun is always non-final in a phrase, and for this reason, the *t* is retained, as it is before a suffix. The fact that *patah* occurs in a verb form before a subject or modifier as it does in a construct noun reflects the fact that verb forms in this position were characteristically closely joined in speech to the following word, much as were construct nouns. For this reason the reflex of *i (and, much more consistently, that of *a) in the final closed syllable of a verb in context corresponds, in general, to its reflex in the same situation in a construct noun, whereas the reflex in the pausal form of a verb corresponds to that in an absolute noun (see Revell 1981, 97–99). For the same reason *nesiga*, which occurs in a word which was closely joined in speech to the following word, occurs in a very high proportion of the possible cases

[27] *Nesiga* is a terminal feature involving retraction of the stress of the penultimate word in a unit where it is closely joined to the final word, see Revell 1987a, §1.18. The use of *qames* on conjunctive *waw* is also a terminal feature. See Revell 1985b, §11, and, for a more detailed presentation of the view given here, §§5–13 there.

[28] See Revell 1985b, §§13, 26.1. The difficulty of describing this phenomenon arises partly from the fact that these units and sub-units cannot be fully defined either in terms of syntactic structure or of accentuation. They are formed from one or more syntactic units according to their semantic relationship. See Revell 1985b, §25.1. An initial approach to objective description of the significant relationships is made in Revell 1987a, §2.1–5.

[29] In the cases in which the forms called "medial" and "terminal" differ in both features, an intermediate form occurs, stressed as terminal but vowelled as medial, as *ʾ'at:ɔ, ḥ'at:û*.

in construct nouns and verb forms, but is proportionately much less common in other forms.[30]

22.1 The cases in which the reflex of *i in an originally open syllable which is closed and final in MT is *patah* in medial position, *sere* in terminal, are as follows.

(i) Where *patah* occurs in such a syllable in the contextual form of a verb, *sere* occurs in the pausal, as *pit:'ah/pit:'eah, yoʾk'al/yoʾk'el*.

(ii) Where *patah* occurs in such a syllable in a construct form, *sere* occurs in the corresponding absolute form, even before a pharyngal, as *mizb'ah/mizb'eah*.

(iii) *Patah* occurs in *piᶜel* infinitive construct forms from roots III*h* and IIIᶜ in context, where the form is followed by a one-word nominal modifier (which may be introduced by the sign of the definite object *ʾet*). *Sere* occurs even in this situation in forms from PTH (2 Chron 2:6) and from ŠLH (Exod 10:4). *Sere* also occurs in forms from roots III*h* where the modifier is longer[31] (Judg 16:22, 1 Kgs 12:32, Ps 102:21, Ezra 3:8, 9, 1 Chron 23:4, 2 Chron 2:13), and in all forms in pausal situations (BLᶜ Lam 2:8, ŠLH Exod 7:27, 9:2, NSH 1 Chron 15:21, 2 Chron 34:12).

(iv) *Patah* occurs in the first of a pair of verbs acting as a semantic unit in *tiš:ɔb'ar wətišk'ab* (Ezek 32:28) whereas *sere* appears in the final syllable of *nifᶜal* imperfect forms elsewhere. The first member of a pair of this sort commonly occurs in medial form.[32]

(v) Other forms which regularly show *sere* in all positions may also show *patah* as a reflex of *i where the syllable is unstressed with *maqqef* or through *nesiga*, both of which reflect the close-joining of that word to the following in speech (see §9.3). The *segol* which is the usual reflex of *i in this situation (see §9) represents a further degree of change towards *a* than does *sere* (see §§5, 37).

22.2 In the case in (i), the medial form is used within speech units (often co-terminous with clauses). The terminal form is used at the end of such units. In (ii), the medial form is used only in the construct phrase, where the noun is closely joined to what follows, and so only within sub-units of speech. The terminal form is used elsewhere. In (iii) the medial form is used only within clauses, under conditions which suggest that the infinitive and its following subject or object were closely joined in speech. In (iv) and (v) the medial form is used only where the word in question is exceptionally closely related to what follows, and so has a range considerably more restricted than that of the construct form. For reasons which are not clear, the degree of relationship to the following word required for the development of *patah* in a final originally open syllable in imperfect or participial forms is much greater (in most cases) than in perfect forms. Nevertheless here too *patah* clearly arises from such joining. There are a few cases in which it is not clear that *patah* and *sere* arise from the use of the form in medial and terminal position respectively,[33] but these do not affect the validity of the conclusion that, in general, the occurrence of *patah* as reflex of *i in an originally open syllable which is closed and final in MT is due to the fact that the word in question was closely joined in speech to what followed, and so acted as non-final in a unit of more than one word.[34]

---

[30] This was noted in Praetorius 1897, 35–36, and is demonstrated in detail in Revell 1987a. However the intonation contours governing a verb and its following subject or modifier differed, at least historically, from those characteristic of the construct phrase, so that *a in a "pre-tonal" open syllable gives rise to *qames* in a noun, but to *shewa* in a verb, and the original *t* of the pronominal affix of 3f.s. perfect verbs is only rarely preserved.

[31] A short modifier would typically form a single subunit with the preceding infinitive; a longer modifier would form a separate subunit. *Nesiga* is similarly rare in a word followed by a modifier of more than one word, see Revell 1987a, §2.3.

[32] See Revell 1981, 88, 92. The syntax in Ezek 32:28 does not require that the two verbs be taken as a semantic unit, but this is perfectly possible, and is indicated by the accentuation. Such close semantic relationship (reflected in speech) is the most likely reason for the appearance of *patah* as reflex of *i in the stressed closed final syllable of the first.

[33] As in the perfect *qal* forms of ʾHB (see §16). In *qɔm'al* (Isa 33:9), the *patah* with *atnah* is anomalous. This may, in fact, be a medial form, reflecting a tradition in which the main division of the verse followed the first clause (which does end with the pausal form *ʾ'ɔrɛs*), separating the general statement made in that clause from the illustrative examples which follow. In other forms showing *patah* as reflex of *i in pause listed in Gesenius-Kautzsch 1910, §29q, the syllable is either originally closed, or the word is a proper noun. Such nouns not infrequently show unexpected vocalization, as the *segol* in *yizrəᶜ'ɛʾl*, and in *-ᶜ'ɛzɛr* where used as the final element in a name.

[34] Conditioning of vowel pattern on a semantic/rhythmic rather than a syntactic basis is also found in nouns of form *CiCC (where the word-initial position of the doubly closed syllable inhibits change in *i in "close-joined" positions, see Revell 1985a, §6.4), and in f.s. *qal* participles ending in -ɔ. In the latter, terminal vocalization, with *sere* in the penultimate syllable, occurs where the word ends a clause, or where it acts syntactically as a noun. Medial vocalization, with *shewa*

23.1 The reflex of *i in syllables of this sort in participles does not follow the same pattern as in other nouns. *Patah* is the usual reflex of *i in an originally open syllable, closed in MT, in construct forms of substantival nouns of two syllables.[35] In *qal* and *pi^cel* participles, however, the reflex of *i in this situation is almost invariably *ṣere* in all positions. Consequently there is no formal mark of the construct state in m.s. forms, although it can be assumed that some forms are construct on the basis of parallels in which plural forms occur, as *š'olep ḥ'ɔrɛb* (Judg 20:35) cf. *šoləp'e ḥ'ɔrɛb* (Judg 20:35). *Patah* occurs in these forms only:

(i) In m.s. *qal* participles before *^cayin* in RG[c], RQ[c] (where the vowel is stressed) and in NT[c] and ŠS[c] (where it is unstressed through *nesiga*). However *sere* occurs in forms from seventeen other roots III[c], and from sixteen roots III*h*, including cases unstressed through *nesiga*.

(ii) Elsewhere only in *^ɔob'ad* (m.s. *qal*, Deut 32:28), *məqarq'ar* (m.s. *pi^cel*, Isa 22:5). The syllable is stressed in both cases.

23.2 A further aspect of the almost complete dominance of the terminal form in m.s. *qal* participles is seen in the fact that the reflex of *i in a final closed syllable unstressed through *nesiga* is *ṣere* in 30 of 31 cases in these participles (97%),[36] as opposed to 22 of 78 cases in verb forms (28%), and 2 of 25 cases in infinitives (8%). In m.s. participles from roots III*h* (except for *^coś'e* and *mak:'e*, where the usage is exceptional) the usual final vowel is *segol*. The use of *ṣere*, producing the form called "construct," appears much less common than in other nouns from these roots, and appears not to be syntactically conditioned. The dominance of the terminal form is, then, characteristic of participles generally. No very convincing explanation for the phenomenon can be suggested, but, since it is not limited to the type of syllable considered here, it alone cannot indicate that the above reasoning about the reflex of *i in the construct forms of other nouns is faulty.

24. *Medial Position: Originally Closed Syllables.* Where a form ends in an originally closed syllable the effect of relationship to context is much less marked than where the syllable was originally open. The correct conclusion seems to be that the reflex of *i in these forms is mainly determined by the phonetic environment. The relation of the form to its context appears to be of secondary importance. It does, however, tend to induce *patah* in terminal position, and *ṣere* in medial position, as in *mes'ab* and *mes'eb*, words commonly separated in the dictionaries, but presumably both originating in the *hif^cil* participle of SBB. The phonetic context of the *i in the originally closed syllable is moderately unfavorable (see §15.3). *Patah* occurs in *mes'ab* (1 Kgs 6:29), where the word functions as an adverb with no following modifiers. *Sere* occurs in *mes'eb* (Jer 21:4), where the word is predicative, with following modifiers (cf. note 34). *Patah* regularly occurs in an originally closed final syllable in perfect verb forms in pause, as *hep'ar* (Gen 17:14), *het'az* (Isa 18:5). In imperfect forms, *patah* generally occurs in favorable phonetic contexts, as *way:el'ak* (Gen 24:61), but not after a sibilant in *way:eš'eb* (Ruth 4:1), and not where the syllable was doubly closed, as *yɔr'ea^c* (Zeph 1:12, cf. Job 31:26).[37] In nouns, where the syllable was doubly closed in all cases, *patah* occurs in favorable contexts in absolute forms, such as *mor'ag*, and the f.s. participle *mešɔr'at* (1 Kgs 1:15), but not in an unfavorable context before *n* in *mɔg'en*, or before a sibilant in *hɔm'eš*. In verb forms in contextual position, the usual reflex of *i in an originally closed final stressed syllable is *ṣere*, but *patah* may occur in a favorable context in perfect forms, as *hed'aq* (2 Kgs 23:15), *hem'ar* (Ruth 1:20). Where the syllable is not stressed, *patah* or *segol* may occur, as *h'etɛl* (Gen 31:7, with *nesiga*), *yɔṣar* (1 Kgs 8:37, with *maqqef*) and *way:'ɔṣar* (2 Chron 28:20). In construct noun forms, the reflex is *ṣere* in *mɔg'en*, *hăm'eš*, and in similarly unfavorable environments in nouns of form *CiC:, as *^ɔem*, and *CiCC, as *š'ekɛl* (see note 21, and Revell 1985a, §6.4). In favorable environments the reflex is *patah* in *CiC:, as *bat*, *segol* in *CiCC, as *b'ɛṭen*. *Patah* also occurs in the un-

in the penultimate syllable, occurs where the word stands within a clause, is predicative or attributive, and is followed by a modifier. See Garr 1987, 144–45.

[35] This is usual in nouns of form *CaCiC (as described), and regular in nouns in which the first syllable was originally closed, as *misp'ad*, *ma^căśar*. In *mapt'eaḥ* (Isa 22:22) the appearance of *sere* in a construct form must be considered an anomaly.

[36] In four of these cases of *nesiga*, *sere* (with *patah* furtive) is retained before *ḥet* (Job 5:10) or *^cayin* (Isa 63:12; Prov 1:19, 11:26).

[37] The use of *patah* cannot be wholly explained on the basis of phonetic context. Thus in *nif^cal* imperfect forms, *patah* occurs in HNQ (2 Sam 17:23) and NPŠ (Exod 31:17) where the context is not particularly favorable, but not in MLṬ (1 Sam 19:12, 17) or SBR (Jer 51:8) where it is relatively favorable.

stressed *CiC: form *qan* (construct, Deut 22:6) despite the unfavorable phonetic environment.

25. *Relation to Context: Conclusions.* It can be concluded, then, that the conditioning ascribed to "relation to context" was effected by the prosodic features characteristic of the units of speech into which the text was divided in its oral presentation, and the subdivisions of these units. Position in the stressed syllable of a word terminal in such a unit is clearly conducive to the change from *i to *patah* where the syllable was originally closed. These units were divided into subunits composed of single words, or of larger syntactic or semantic structures. Terminal position in such a subunit has an effect similar to that of terminal position in the larger unit, but less marked. Final position in a word which is medial in a subunit has the same effect as internal position in a word, where the syntactic or other semantic relations between the words in the subunit led to their being read as a closely-knit unit. This was no doubt characteristic of the construct phrase, but must also have been usual for a verb with following subject or modifier.

OTHER CONDITIONING FACTORS

26. The above information includes two rather striking anomalies: the common use of *ṣere* in a closed stressed non-final syllable in feminine plural imperfect forms, and the fact that, in any stem, *patah* is much more common as a reflex of *i in perfect forms than in imperfect. These anomalies could be explained by the assumption that vowel change might be affected by the commonness of a word in use, or by its length. If these are conditioning factors, however, their effect is marginal, and cannot be clearly demonstrated.

27. *Patah* occurs as the reflex of *i in some f.pl. imperfect and imperative forms, but *ṣere* is a much more common reflex in the closed stressed non-final syllable of these forms than in any other form in which the syllable is neither word-initial nor closed by a lengthened consonant (see §6.2.iii).[38] It is noteworthy that the reflex of *a in these forms is regularly *pataḥ* in pausal position, not the expected *qameṣ*. *Pataḥ* occurs in *tišp'alnɔ* (Isa 5:15, with *silluq*), and in forms from ḤRŠ (Mic 7:16), YṢT (Jer 49:2), MWG (*hitpolel* Amos 9:13), PRḤ (Isa 66:14, before *ḥet*), QŠB (Isa 32:3), and ŠBᶜ (Prov 27:20, 30:15, before ᶜ*ayin*!). The only example showing *qameṣ* I have noted is *tīšɔmnɔ* (Ezek 6:6, with *atnaḥ*). Where the subject is feminine plural, imperfect verbs typically appear in the (originally) masculine forms at Qumran, although the distinctive feminine form is still usually used for imperatives (see Qimron 1976, §§310.128, 310.135). In Mishnaic Hebrew, feminine plural forms are not used in either imperfect or imperative. In the later stages of the development of the reading tradition, then, these feminine forms only existed in the few examples in the biblical text. It could well be this rarity which has inhibited the expected development of the vowels in their stressed syllables.[39]

28.1 *Pataḥ* is a common reflex of *i in a closed final stressed syllable in perfect verb forms, even where the vowel is not followed by *ḥet* or ᶜ*ayin*, but this is not the case in imperfects. In *qal* forms, in a closed final stressed syllable (originally open) in context, perfect forms from 13 roots show *pataḥ*; those from 6, *ṣere*, and those from 3, both vowels. In imperfect forms under the same conditions, forms from 3 roots show *pataḥ* (the preceding syllable has *ḥolem* in all cases, see §19.2). Forms from seven roots show *ṣere* (the preceding syllable has *ḥolem* in two cases). *Pataḥ* is somewhat less common in perfect *piᶜel* forms than in perfect *qal*, but only occurs in one imperfect form (see §8.2.i.b). In *hifᶜil* forms from geminate roots (where the closed stressed final syllable was originally closed), *pataḥ* occurs in this final syllable in perfect forms from DQQ, MRR, PRR (only in pause), ṢRR, QLL, RKK, TZZ (in pause). *Ṣere* occurs in forms from HLL, SBB, ᵓZZ, PRR (only within a clause).[40] *Ṣere* occurs in all imperfect and imperative forms under the conditions given (forms from HLL, ḤLL, YLL, MŠŠ, NSS, SBB, SKK, PRR, QLL, TMM). *Ṣere* also occurs in imperfect forms from Rᶜᶜ in pause, *pataḥ* only in forms from Rᶜᶜ within a clause. There is thus a consistent difference in the development of *i in the same type of syllable in perfect and imperfect forms.

28.2 Difference in development between perfect and imperfect forms is also found in the *qal* of hollow and

[38] On the possibility that *ṣere* in these forms results from analogic change, see §34.3. On the fact that *pataḥ* occurs regularly in the stressed syllable of f.pl. *nifᶜal* imperfect forms, see §19.2.

[39] Where vowel change is caused by the process of articulation (as is suggested for these changes in §30), the more frequently the process occurs, the greater should be the change. See the hypothesis formulated in Phillips 1984, 336: "Physiologically motivated sound changes affect the most frequent words first."

[40] Anomalous *ḥireq* occurs in the *hifᶜil* perfect of PRR in Ezek 17:19 (in pause) and Ps 33:10 (within a clause).

geminate roots. Perfect (and imperative) forms from the *qal* stem of these roots quite commonly show final stress in forms composed of two syllables of which the first is long and the second open (as in 3f.s. and 3p.pl. perfect forms such as *bɔzʼɔ*, 2 Kgs 19:21, *qal:ʼū*, Jer 4:13). Final stress is extremely rare in imperfect forms, and in forms from other stems, all of which are of more than two syllables. The determining difference appears, then, to be the length of the form.[41] This difference in length presumably gives rise to some difference in the prosodic patterns characteristic of the forms, and such a difference might also affect the forms discussed here, resulting, for instance, in the strange fact that the final syllable of the *hifᶜil* forms from geminate roots acts as originally closed in perfect forms (*pataḥ* in pause), but acts as originally open in imperfect forms (*ṣere* in pause). The perfect forms in question are all of two syllables. The imperfect forms are longer in *qal* and *piᶜel*, and were originally longer in *hifᶜil* (*yVhaCiC: > *yɔCeC*). This original difference in structure is presumably the cause of the retention of *pataḥ* in the prefix of imperfect *hifᶜil* forms, as against the development of *ḥireq* in *hifᶜil* perfect and *qal* imperfect forms (*yVhaqtil > *yaqtîl*, but *haqtil > *hiqtîl*, *yaqtul > *yiqtol* and *yantin > *yit:en*). The same difference of original structure might, then, account for the different development of *i in a final closed syllable in the perfect and imperfect forms from the *hifᶜil* of geminate roots. It should be noted, though, that imperative forms, which are also of two syllables, generally show *ṣere*, as do imperfects, and only rarely show *pataḥ* like perfects.[42]

29. The MT reflex of *i is, in most cases, easily explained as the result of natural sound change conditioned by the phonetic environment of the vowel, the structure and position of the syllable in which it stands, and the relation of the word in question to its context. A high proportion of unexpected reflexes appear in two morphologically defined groups. The explanation offered here is, perhaps, a probability for the one, but is no more than a possibility for the other. The existence of such groups, along with that of the various individual cases of unexpected reflexes discussed above, might seem to support the view that the MT reflexes of *i in general do not result from natural conditioning, which should cause the same change in the same sound under the same conditions. Due to the nature of spoken language, however, it is extremely difficult to determine when the conditions can justifiably be described as "the same." This problem is compounded when the language is accessible only through written sources. The level of irregularity shown here does not present a serious challenge to the view that, in general, the MT reflex of *i results from natural sound change conditioned as stated at the beginning of this paragraph.

HISTORICAL CONSIDERATIONS

30. *The Process of Change.* The above description of the development of *i suggests that conditioning factors of two types are mainly involved. These are articulatory factors, described in terms of the preceding and following sounds—usually consonants, occasionally vowels—and prosodic factors, described in terms of the structure of the syllable, and its position in the word or longer sub-unit of speech.

30.1 The conditioning of sound change by articulatory factors results from the fact that spoken language is a continuum both in articulatory and in acoustic terms. While one organ (as the tongue) may be momentarily still, others (as the lips) will be in motion. As a result, the phonetic characteristics of neighboring sounds are inevitably intermingled in the passage from the peak of one syllable to that of the next, so that each sound is affected to some extent by its neighbors. In the case of "vowel harmony," the peak of one syllable affects that of its neighbor in a similar way. This sometimes seems to be involved in the development of *i (see § 19.2), but consonantal sounds obviously have a greater effect. In the case of a preceding consonant, the manner of articulation appears to be the most significant conditioning factor. Voiced or stopped sounds appear to be most conducive to the change of *i to *pataḥ* (see § 18.1), but possibly some concomitant feature, common to sounds of both types—perhaps lax or open articulation—was the real cause of change. In the case of a consonant following *i, the most significant factor seems to have been the position of articulation. The change to *pataḥ* is induced by consonants articulated at the back of the mouth, presumably because, during the realization of the vowel, the tongue anticipated the backward position required for

[41] The special case of perfect forms with *waw* consecutive is not included in this statement. The development of final stress in the verb forms in question is discussed in Revell 1987b.

[42] There are three examples of *piᶜel* imperatives showing *pataḥ* in a final syllable closed by a consonant other than *ḥet* or *ᶜayin* (Ezek 37:17, Ps 55:10, and, with *maqqef* Job 36:2) compared to nearly fifty with *ṣere*. *Pataḥ* never occurs as a reflex of *i in *qal* imperatives (where the syllable is always word-initial, as *šeb*), or in *hifᶜil* forms from geminate roots.

the articulation of the following consonant. Conversely the change is inhibited by consonants articulated at the front of the mouth, particularly by *nun*, for which the tongue position must have been very close to that required for *hireq*.

30.2 The source of the conditioning of sound change by prosodic factors is more difficult to establish. It is evident that a closed syllable was favorable to the change from *i to *patah*, whereas an open one was not. Internal position was favorable to the change in a closed syllable; final position was not. Evidently the articulatory factors had maximum effect on the vowel of a closed internal syllable. Vowel change was also greater where *i was followed by two different consonants than where it was followed by one, even if that one consonant was lengthened. Evidently where the vocal organs had to assume two different positions between the peaks of neighboring syllables, the interference with the vowel of the first was greater than where they had to assume only one, even if that one position was held to produce a long consonant.[43] It seems reasonable to suggest that the immediate conditioning factor was the length of the vowel. Interference from neighboring sounds would affect a greater proportion of the realization of a vowel if it were short than if it were long, and would thus have a greater effect on the auditory perception of it. It is reasonable to suggest that the realization of the reflex of an originally short vowel was shorter than average in an internal closed syllable, and longer in a final closed syllable (see now Kahn 1987, 33, 44); shorter where the vowel was followed by two different consonants than where it was followed by one lengthened one. The change from *i to *pataḥ* was also inhibited where the syllable was word-initial. Since the reflex of *a is frequently *hireq* in a closed initial syllable, it seems probable that some prosodic feature tended to induce a high front vowel in such syllables generally, so that this is not a feature characteristic of the development of *i alone.

30.3 It has been assumed throughout this discussion that the change from *i to *patah* was a slow process. Where articulatory or prosodic factors were favorable to change, the process would occur more quickly than where they were not, resulting in different MT reflexes. It seems probable that a difference in MT reflexes also resulted from the fact that the process took place over a longer period in some situations than in others. It was argued in §21 that this is the reason that change is characteristically greater in terminal (pausal) position than in medial. Possibly it is correct to say that the change affected closed stressed internal syllables in words in terminal position first, then other closed stressed internal syllables, then closed stressed final syllables, and so on according to stress level. It was further suggested in §21 that the treatment of final syllables also reflects a time factor. Where the word is in terminal position, *i in an originally closed stressed final syllable acts as it does in a closed stressed internal syllable. Where the word is in medial position, *i in such a syllable acts much as it does in an originally open syllable. Evidently the factor responsible for this latter situation, suggested as the lengthening of the vowel in §30.2, became effective after the change was well advanced in terminal position, but before it had progressed far in medial position. As suggested, this lengthening would inhibit or prevent further progress of the change from *i to *patah*, which occurs under stress only where the vowel is short. Such lengthening of originally short vowels probably affected stressed open syllables first, and affected closed syllables only under stress, first in final position, then in internal position.

31. Consequently it is suggested that the MT reflexes of *i studied here result from two processes: the change from *i towards *a* (or ɔ), and the "lengthening" of the reflex of *i. These two processes continued for some time more or less concurrently, but were completely independent. The change towards *a* had its greatest effect in closed internal syllables, with a lesser effect in closed syllables elsewhere. Lengthening had its greatest effect in open syllables, including the originally open syllables considered here, but also affected originally closed syllables under stress. Both processes had their greatest effect under pausal stress, and a lesser effect under stress elsewhere. The change took place only in short vowels. Once the vowel had been lengthened, the change effectively ceased. The progress of these two processes is described below in terms of "stages," which are theoretically descriptive of the situation at a point in time. The processes proceeded continuously, however, and were conditioned by a variety of factors, so it is wrong to think of each stage as neatly completed before the next began.

31.1 The change first affected *i in closed internal syllables under pausal stress, eventually giving rise

[43] Cf. the use of *qibbus* reflecting original short *u before a lengthened consonant in a closed unstressed syllable, as *huq:'ī* as compared to the lower *qames* where the vowel is followed by two consonants, as *qɔdš'ī*.

to *a*. The typical MT reflex is *pataḥ*, but under particularly favorable conditions, the change proceeded fast enough that the resulting *a* gave rise to *qameṣ* (the usual reflex of original *a in pause) before the process was stopped.

31.2 The change next began to affect other closed stressed internal syllables, and also originally closed stressed final syllables in pause. The usual MT reflex is *pataḥ*. Since the change in these syllables began later than in those described in §31.1, the change to *pataḥ* was often not completed, in which case the reflex is *ṣere*. For the same reason the change never progressed so far that the MT reflex is *qameṣ*.

31.3 Lengthening then began to affect closed syllables in pause. In originally open syllables, the change from *i to *a* had had little effect, so that the MT reflex is regularly *ṣere*. In originally closed final syllables, the change had had greater effect, so that the MT reflex is usually *pataḥ*. In closed internal syllables, the MT reflex is *qameṣ* where the situation is favorable to the change from *a* to *qameṣ* (see §19.3). Where the conditions were not favorable to this change, further development was inhibited by the lengthening of the vowel, and the MT reflex is usually *pataḥ*.

31.4 Lengthening subsequently affected stressed closed syllables elsewhere. The steady progress of the change in closed internal syllables (including syllables final in a word, but internal in a closely-knit unit of two or more words) gave rise to MT *pataḥ*, except in situations unfavorable to the change. In word-final syllables which are not internal in a larger unit, the progress of the change had been much slower, and had much the same effect on originally closed syllables as on originally open ones. Consequently the MT reflex of *i is *ṣere* in syllables of both types which are not in pause, except under conditions highly favorable to change.

31.5 At some point in this process, the change also began to affect closed unstressed syllables, and was not inhibited by the lengthening of the vowel. Consequently the MT reflex of *i in these syllables is usually *segol* or *pataḥ*, rarely *ṣere* or *ḥireq*.

32. *The Development of* Segol. The above outline gives a clear idea of the relationship of *ṣere*, *pataḥ*, and *qameṣ* as reflex of *i. The position of *segol* is more complex. In an open syllable, *segol* is the reflex of *a or *i in an originally closed syllable which has been secondarily opened through segolation or the effects of a following guttural. *Segol* reflecting *i is generally presumed to represent a stage in the change from *i towards *a*, as explained above. Where this change has been completed, however, the resultant *a* may continue to develop as does *a. Where *a in an open syllable is not reduced to *shewa*, it typically gives rise to *qameṣ* where the conditions are favorable to this change. Where conditions are less favorable, it gives rise to *segol*. Thus in nouns of form *CaCC, the reflex of *a is *qameṣ* in terminal position, *segol* in medial position. In an unstressed syllable before a historically lengthened guttural followed by *qameṣ*, the situation is rendered less favorable to the change (giving rise to MT *segol* rather than *qameṣ*) by a following consonant which is articulated further towards the front, or is voiceless (and so *hɛḥ'ɔg*, but *hɔh'ɔr*, *hɔᶜ'ɔm*), or by a following vowel which is unstressed (and so *hɛᶜɔr'îm* but *hɔᶜ'ɔm*). The process of change was also apparently inhibited where the following vowel stood in a closed unstressed syllable, so the reflex of *a is *pataḥ* in *haḥɔkm'ɔ*.[44] Apparently, then, the low central vowel *a (or *a*) shifted to mid back under favorable conditions, and so was perceived as *qameṣ*. Under less favorable conditions the full process of change was not completed, and the vowel was perceived as the same sound as the mid central vowel (*segol*) which arose as a stage on the shift from *i towards *a*. Thus the MT reflexes of *a (or *a* reflecting *i) can be explained as resulting from a process similar to that suggested for *i, and the same appears to be true for *u (see nn. 43, 52). A process of change was conditioned by a variety of factors, giving rise to different MT reflexes according to the degree to which the interplay of the conditioning factors provided an environment favorable or unfavorable to change in different situations.

[44] If the stressed *segol* in *h'ɛrɔ n:'ɔsû* (Gen 14:10) represents the same phenomenon, the change in the vowel is probably accounted for by the stress on the vowel, and the following voiced *r*. The above view sees the change to *segol* as (partial) vowel harmony (as suggested by Joüon 1947, §§29f). The required assumption that the *ḥaṭef shewa* of forms like *hɛḥŏdɔš'îm* had a greater effect on preceding *a than did the vowel of a closed unstressed syllable is partially explained by the finding of Kahn 1987, 39, that a *ḥaṭef* or vocal *shewa* was no shorter than the vowel of a closed unstressed syllable. The more generally accepted view, that *segol* arose from dissimilation before *ā*, and so not before the reflex of *u in *haḥɔkm'ɔ* fails to explain the occurrence of *segol* before the reflex of *u in forms like *hɛḥŏdɔš'îm* (as remarked in Blau 1977, note 14) and also does not explain why this dissimilation was restricted to such a limited range of situations.

33.1 In closed syllables, *segol* occurs as the reflex of *i under stress in a few cases. In the perfect *piʿel* forms of DBR, KBS, and KPR, the favorable features in the phonetic environment presumably induced the change from *i towards *a*, but the unfavorable features prevented more than a partial development of it (see §§14.2, 14.4). The 2p.pl. and 3p.pl. pronominal elements *-t'ɛm*, *-k'ɛm*, *-h'ɛm*, etc., presumably reflect an original form *-C'iC:V. The expected reflex of *i in a closed internal stressed syllable would be *patah*, but the following lengthened consonant would be unfavorable to this development (see §6). Consequently here also the *segol* probably results from a change of *i towards *a* which has not been completed. However the consistent vowelling of these morphs in MT no doubt results, to some extent, from levelling.[45]

33.2 *Segol* before 2m.s. or 2p.pl. pronominal suffixes, in a syllable originally open, but closed in MT (see §10), must also represent a partial change from *i towards *a*. The fact that the syllable is unstressed no doubt contributed to the failure to complete the change, but this would not account for the appearance of *hireq* in this situation (see §6.1). The fact that the process of change could only begin after the syllable became closed, a relatively recent development, must be a contributory factor.

33.3 In unstressed closed final syllables, *segol* no doubt also results from a change from *i towards *a* which has not been completed.[46] Some verb forms—imperfects with *waw* consecutive, some jussives, and an imperative—regularly end in closed unstressed syllables in contextual position, although stress is final in pause. A final syllable which is regularly stressed in contextual position may loose its stress because the word has *maqqef*, or because stress is retracted through *nesiga*. In these two latter situations, *segol* undoubtedly results from a change from *i towards *a* induced by the fact that the syllable was internal in a unit of several words (see §22). The change was not completed because the syllable was unstressed, and/or because the conditions inducing the change were of recent development. There would be no need to consider the case of *segol* in the closed unstressed final syllable of a verb form in context separately, were it not for the common opinion that, in these forms, the original stress position was penultimate.

34. The opinion that penultimate stress is original in these forms is supported only by the assumption that the existing facts cannot otherwise easily be explained (so Blau 1981, §1.1), a view that can certainly be contested (see Revell 1984, §13). In the present context, this opinion presents several problems.

34.1 There is no reason to suppose that *segol* developed in the final syllable of these forms would change when the syllable became stressed through a secondary development. *Segol* remains in a closed stressed final syllable elsewhere (see §33.1).

34.2 The *patah* which usually develops in the final closed syllable of *waw* consecutive imperfect forms under pausal stress is the expected reflex of *i, and the lack of consistency in the development of *patah* in this position is also expected (see §7). The description above suggests that it is extremely unlikely that *patah* would have developed in so many forms if final stressing were a secondary development, unless it were an extremely early one.

34.3 Proponents of the view that final stress in these forms is secondary generally argue that *sere* in a stressed final syllable in any *waw* consecutive imperfect form must be due to analogic change. But *sere* is the expected reflex of *i in an originally stressed final syllable of an imperfect form which is closed in MT, whether originally closed or originally open. Moreover in the case of the *hifʿil*, the imperfect paradigm develops *sere* in final syllables only in those syllables in which *sere* is supposedly due to analogic change, in

[45] The attested Semitic forms suggest that the immediate ancestors of these pronominal morphs had *u* or *i* as the first vowel, and *m* or *n* as the following consonant, with either the vowel or the consonant (but not both) long, and either the vowel or the consonant (but not both) used to distinguish the genders. The evidence suggests that Hebrew used a consistent set of choices from these possibilities (as did Akkadian and Arabic, not an inconsistent set, as Aramaic). If so, the immediate ancestor of the Hebrew morphs had the form *CiC:V. The use of *segol* in the MT forms in pause (where *patah* might be expected, see §24) and in feminine forms (where *sere* would be expected, since the following *nun* provides an unfavorable phonetic environment, see §18.1) certainly suggests some levelling. Note, however, that *sere* occurs in the independent forms *hem*, *h'em:ɔ*, etc., where the situation is particularly unfavorable of change due to the word-initial position of the vowel (see §6) and the laryngal preceding it (see §16).

[46] *Patah* occasionally occurs in these situations (see §9.3) as does *sere* (see §§9.2, 23.2), showing that the usual view that *segol* here represents *sere* shortened due to loss of stress is untenable.

waw consecutive or jussive forms. The other forms of the imperfect do not provide a model for this change. The same is true for *sere* in an originally closed stressed syllable in forms such as *tɔg'elnɔ*, *ha'z'en:ɔ*.

34.4 Even where there are related forms which can be suggested as the model for the analogic change which supposedly produced the MT forms, no single trend of change appears, showing assimilation to a dominant pattern. Thus the appearance of *patah* in the closed unstressed final syllable of *way:'o'kal* (a serious problem if stress were originally penultimate) is said in Blau 1981, §4, to be due "to analogy, *inter alia* of the pausal form." Analogy in this case, then, affected vowel pattern but not stress position. Yet in *way:in:ɔp'eš* analogy (not with the pausal form of the same word, but with that of the imperfect indicative) is said to have caused change in stress position.[47]

35. If it is assumed that stress was originally final in these forms, but has been retracted in a small group of them,[48] their vowelling can be shown to be consistent with the patterns described above. Where the syllable was closed, final, and stressed, but not in pause, *i would normally give rise to *sere* except under conditions exceptionally favorable to the development of *a*. Consequently *patah* appears only before *ḥet* and *ʿayin*, and also in forms of the *yo'k'al* type for the reasons given in §19.2. Where the stress was retracted, this *patah* naturally remains unchanged. Elsewhere where stress was retracted, the vowel is *segol*, save in one form in which *hireq* appears, and one in which *sere* is used (see §§9.1, 2. Both occur before *sade*, a phonetic environment unfavorable to change, see §18.1). This *segol* represents an incomplete change from *i to *a*, just as does *segol* in syllables unstressed through *maqqef* or *nesiga*. The process of change was not inhibited through the lengthening of the vowel, as it was where the syllable was stressed, but the stress level was not adequate to induce its completion.

CONCLUSION

36. The most significant innovation in this paper is the recognition of the importance of the phonetic environment of the vowel as a factor conditioning the reflex of *i. Since many different factors were involved in this development, a clear description of the individual significance of any one of them is impossible. Nevertheless the attempt to describe the part played by these various factors in conditioning the development of *i makes it quite clear that the MT reflexes developed, for the most part, through natural sound change. Interference form analogic levelling appears to have been slight. A further significant result of this study is the demonstration of the limited effect of vowel length in the development of the different reflexes of originally short vowels. The correlation of different vowel qualities with different categories of length, which remains typical of grammars, represents the modification of an ostensibly synchronic description by historical considerations. This paper suggests that such considerations, whether they relate to the earlier or to the later history of the language, have inhibited the just evaluation of the internal information on the development of originally short vowels available in MT.

[47] See Blau 1981, 2, n.4, and cf. p. 10, §4.1.4, where the imperfect indicative form is said to determine the vowelling of the imperative.

[48] In the writer's opinion, penultimate stress in these verb forms with a closed final syllable results from particularly close relationship to what follows, just as does final stress in words with an open final syllable (see §20). This is the reason why final stress occurs in forms like *way:es'e'*. However this is a subject for a later paper.

BIBLIOGRAPHY

Ben Hayyim, Z. 1978. Thoughts on the Hebrew Vowel System. *Studies in the Bible and the ancient Near East.* Pp. 95–105. Jerusalem: A. Rubinstein.

Blau, J. 1977. An Internal Proof for the Relative Chronology of Tiberian *qames* (= "Marginalia Semitica III.1"). *Israel Oriental Studies* 7: 14–17.

———. 1981. On Pausal Lengthening, Pausal Stress Shift, Philippi's Law, and Rule Ordering in Biblical Hebrew. *Hebrew Annual Review* 5: 1–13.

Bauer, H. and Leander, P. 1922. *Historische Grammatik der hebräischen Sprache.* Halle.

Díaz-Esteban, F. 1975. *Sefer 'Oklah We-'Oklah.* Textos y Estudios, vol. 4. Madrid.

Driver, S. R. 1892. *A Treatise on the Use of the Tenses in Hebrew*, 3rd ed. Oxford.

Eldar, I. 1985. Torat ha-Qeri'a ba-Miqra. *Tarbis* 54: 225–43.

Garr, W. R. 1987. Pretonic Vowels in Hebrew. *Vetus Testamentum* 37: 129–53.

Gesenius-Kautzsch. 1910. *Gesenius' Hebrew Grammar*. Ed. E. Kautzsch. Tr. A. E. Cowley. Oxford.

Joüon, P. 1947. *Grammaire de l'hébreu biblique*, 2nd ed. Rome.

Kahn, G. 1987. Vowel Length and Syllable Structure in the Tiberian Tradition of Biblical Hebrew. *Journal of Semitic Studies* 32: 23–82.

Lambdin, T. O. 1985. Philippi's Law Reconsidered. In *Biblical and Related Studies Presented to Samuel Iwry*, ed. A. Kort and S. Morschauser. Pp. 135–45. Winona Lake, Ind.

Levy, K. 1936. *Zur masoretischen Grammatik*. Bonner Orientalistische Studien, Nr. 15. Stuttgart.

Merx, A. 1889. *Historia artis grammaticae apud Syros*. Abhandlungen für die Kunde des Morgenlandes IX.2. Leipzig.

Moscati, S. 1964. *An Introduction to the Comparative Grammar of the Semitic Languages*. Wiesbaden.

Phillips, Betty S. 1984. Word frequency and the Actuation of Sound Change. *Language* 60: 320–42.

Praetorius, F. 1897. *Über den rückweichenden Accent im Hebräischen*. Halle a. S.

Qimron, E. 1976. *A Grammar of the Hebrew Language of the Dead Sea Scrolls*. Jerusalem. (A new edition is now translated as *The Hebrew of the Dead Sea Scrolls* [Atlanta, 1986].)

Rabin, C. 1968. The Vocalization of the Third Singular Perfect of *Piᶜēl* in Tiberian Hebrew. *Leshonenu* 32: 12–26.

Revell, E. J. 1981. Syntactic/Semantic Structure and the Reflexes of Original Short *a* in Tiberian Pointing. *Hebrew Annual Review* 5: 75–100.

———. 1984. Stress and the *Waw* Consecutive in Biblical Hebrew. *Journal of the American Oriental Society* 104: 437–44.

———. 1985a. The Vowelling of "*i* type" Segolates in Tiberian Hebrew. *Journal of Near Eastern Studies* 44: 319–28.

———. 1985b. The Conditioning of Stress Position in *Waw* Consecutive Perfect Forms in Biblical Hebrew. *Hebrew Annual Review* 9: 277–300.

———. 1986. Pausal Phenomena and the Biblical tradition in Texts with Palestinian Pointing. In *Salvacion en la palabra: Targum-Derash-Berith, en memoria del Professor A. Diez-Macho*, ed. D. Muñoz-Leon. Pp. 235–44. Madrid.

———. 1987a. *Nesiga in Tiberian Hebrew*. Textos y Estudios, vol. 39. Madrid.

———. 1987b. Stress Position in Verb Forms with Vocalic Affix. *Journal of Semitic Studies* 32: 249–71.

Skoss, S. L. 1952. A Study of Hebrew Vowels from Saadia Gaon's Grammatical Work 'Kutub al-Lughah'. *Jewish Quarterly Review* 42: 283–317.

Yeivin, I. 1980. *Introduction to the Tiberian Masorah*. Missoula.

———. 1985. *The Hebrew Language Tradition as Reflected in the Babylonian Vocalization*. Jerusalem.